

International Trade Responses to Labor Market Regulations[†]

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This paper demonstrates that labor market regulations, such as minimum wages or payroll taxes, shape trade competition in labor-intensive activities. I exploit data from a large European trade program where firms from different countries supply labor services at the same location but face different payroll taxes and minimum wage rules. Country case studies and model-consistent gravity estimates show large trade responses to tax and regulatory reforms, with an elasticity of trade in services to labor costs larger than one. The results imply that absent regulatory and fiscal harmonization, export competitiveness depends, in part, on domestic labor market policies. (JEL F14, F16, F42, H25, J22, J23, K31)

It is a well-known theoretical result that trade is mutually beneficial because countries specialize in activities where they have a relative cost advantage. In public discourse, policymakers often emphasize conventional sources of comparative advantages, such as factor abundance, technology, or economies of scale, leaving more politically controversial explanations aside. For instance, former WTO director Pascal Lamy once argued that lower taxes and weaker social standards are unlikely to govern international trade patterns because “differences in wages largely reflect differences in labor productivity.”¹

In practice, trade economists have long considered institutions as determinants of international trade patterns (Vogel 2007; Costinot 2009; Chor 2010), focusing on financial development (Manova 2008), contract enforcement (Levchenko 2007; Nunn 2007), or labor market rigidities (Helpman and Itskhoki 2010; Cuñat and Melitz 2012). At the same time, labor economists have long studied how labor market institutions affect the structure of earnings and employment *within* countries, often exploiting regional variations in minimum wages or payroll taxes (e.g., Cengiz et al. 2019; Dustmann et al. 2022; Ku, Schönberg, and Schreiner 2020). However, there is still little credible empirical evidence that these labor market institutions can shape trade specialization and international trade patterns.

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¹ This is from a 2010 speech (https://www.wto.org/english/news_e/sppl_e/sppl152_e.htm) by Lamy on “facts and fictions” in international trade.

Yet social standards could represent an entirely distinct form of comparative advantage, in contrast to conventional explanations of global competition. First, people find the idea that international specialization could stem from the distinct regulatory choices made by different countries unfair and unappealing (Di Tella and Rodrik 2020). Second, if labor institutions cause trade, some countries might then be tempted to lower their standards to gain market shares in labor-intensive sectors, leading to inefficiently low levels of labor taxes and regulations (Krueger 1997).² This race to the bottom could stop certain countries from maintaining their preferred levels of redistribution and workers protection, which would lower political support for economic integration (Rodrik 1998).

This paper provides evidence on the effects of payroll taxes and minimum wages—two key and common dimensions of domestic labor market regulations—on trade in labor-intensive services. I exploit a European trade program for labor services that offers three important advantages to study this question.

First, the EU posting policy allows countries to directly trade labor services and labor institutions across borders, akin to the textbook model of trade in tasks proposed by Grossman and Rossi-Hansberg (2008). The policy regulates temporary contracts performed locally by foreign firms. For instance, a Polish construction firm (exporting country) can build in France (importing country) by sending its employees to France. These posted workers maintain their Polish employment contracts but physically work in France, serving the same customers, at the same location, using shared capital and infrastructure, all while operating under different employment laws. This setting keeps most determinants of comparative advantage constant, while labor market regulations vary by origin of the exporting firm. Furthermore, disparities in labor productivity for customer-facing services across countries are often limited (Balassa 1964; Samuelson 1964), making institutional factors plausibly the primary driver of wage determination for those workers.

Second, the European Commission records payroll tax information on workers involved in cross-border service contracts within the European Union. Thus, I have administrative records of trade in labor services and directly observe the number of workers involved in those cross-border transactions from 2009 to 2018.³ I have augmented those datasets with information on wages, payroll taxes, and minimum wages applicable to firms engaged in the program.

Third, I can exploit many sources of variation in both tax policy and minimum wages: (i) payroll taxes and minimum wages vary across countries and over time, (ii) the European Union has imposed destination-based payroll taxation on some exporting firms, and (iii) the European Union has imposed destination-based minimum wages on some exporting firms. Together, these policy changes create large quasi-experimental variation to identify the causal effects of labor market regulations on trade in labor-intensive services within the European Union.

I start by setting out a framework that shows how payroll taxes and minimum wages impact incentives to trade labor services in the European Union. This allows

²Concerns about a “regulatory race to the bottom” and its implications for trade agreements have also been analyzed by Staiger and Sykes (2009) and Grossman, McCalman, and Staiger (2021).

³Those datasets have been previously used by Muñoz (2024) to measure the labor and product market effects of the posting policy.

me to microfound my identification and estimation on a standard Eaton and Kortum (2002) model of trade while exploiting plausibly exogenous sources of identifying variation. Firms compete to produce the same services produced with labor only. As in Ottaviano, Peri, and Wright (2013), foreign workers' wages are contracted in their origin country. Payroll taxes are origin based: Workers posted from different origin countries are subject to different payroll tax reforms, even if they work in the same destination. Minimum wages are destination based: Foreign firms must match the minimum wage abroad during the posting assignment if they pay below that wage floor at home. Within the same origin country, workers posted to different countries are subject to different destination-specific minimum wage reforms. Within the same receiving country, foreign workers posted from different origin countries are either affected or not affected by changes in the local minimum wage.

I then proceed to estimate the elasticity of posting flows to labor costs at three successive levels. First, I explore cross-country correlations between labor cost differentials and trade patterns in Europe. While bilateral trade in goods and services usually responds similarly to gravity forces (Eaton and Kortum 2018), I show a fundamental difference in their sensitivity to labor costs. There is a tight negative correlation between bilateral posting flows and labor cost differentials, with a steady-state elasticity of -0.6 . In contrast, the relationship between trade in goods and labor cost differentials is weakly positive. This is evidence that labor costs drive trade in labor-intensive activities but not trade in less labor-intensive activities. Focusing on services, which primarily rely on labor inputs, is thus more likely to yield a reliable estimate of how sensitive trade flows are to wage costs.

Next, I present the results from the estimation of a theory-based gravity model, using all sources of variation in payroll tax rates and minimum wages across all EU countries from 2009 to 2018. I find large responses of posting flows to changes in labor costs. My estimates of the model-implied elasticity range between -0.7 and -2.1 . The most restrictive specification absorbs origin-year and destination-year fixed effects and yields an elasticity of trade in services with respect to labor costs of -1.2 . I also estimate the effects on other margins of trade in factors: Bilateral trade in goods increases by 0.1 percent, and bilateral migration by 0.2 percent, in response to a 1 percent increase in posting-specific labor costs, but these effects are not always statistically different from 0.

I finally turn to posting responses to two country-specific reforms that shifted (i) minimum wages and (ii) payroll taxes, from origin to destination countries. These reforms were implemented in specific sectors, allowing me to exploit narrower variations in wage costs compared to the gravity estimation. I also use those reforms to test for preexisting trends and to capture the dynamic adjustment of posting flows to changes in labor costs.

The minimum wage reform was introduced in Germany. Until 2015, posted workers operating outside the construction sector had minimum wages set in their origin country while working in Germany. In 2015, those workers became subject to the German minimum wage.⁴ During the same period, posted workers in the construction sector were always subject to German wage floors. After this reform,

⁴Dustmann et al. (2022) previously studied domestic employment responses to this reform.

postings to Germany decreased by 60 percent in the treated sector compared to the construction sector. The reduction in exports to Germany came from low-wage origin countries that had to match the new German minimum wage. In contrast, the effects on postings from high-wage origin countries, where the new minimum wage was not binding, are statistically insignificant.

The payroll tax reform was introduced in Luxembourg. Starting in May 2010, Luxembourgian firms exporting temporary employment agency services abroad through posting had to pay payroll taxes in the destination country during the posting assignment of their workers. Postings from Luxembourgian firms in the exposed sector dropped dramatically just after the reform. In comparison, both the domestic supply of the same services and exports of posting services in a sheltered sector remained unaffected. Overall, the case studies confirm that trade in labor services is highly sensitive to payroll tax rates and minimum wages, with elasticities that are systematically above one. The absence of pre-reform trends in both analyses confirm the exogeneity of those labor cost changes and validate using those reforms as a source of identification.

The main contribution of this paper is to estimate the sensitivity of trade in services to labor costs. I find that heterogeneity in labor market regulations matters for trade patterns in labor-intensive activities, meaning that trade integration can limit the ability for countries to maintain independent labor market rules. In addition to informing the partial equilibrium effects of labor cost shocks, my estimates also inform the general equilibrium effects of trade integration in services. The trade elasticity for labor-intensive services is smaller than the trade elasticity for goods (Head and Mayer 2014), which has implications for the welfare analysis of trade policies in those sectors (Arkolakis, Costinot, and Rodríguez-Clare 2012).

This result contributes to a broad literature investigating institutional sources of international specialization (Vogel 2007; Levchenko 2007; Manova 2008; Chor 2010) and the role of factor intensity on trade patterns (Romalis 2004). I study the role of labor market institutions for international trade.⁵ This question is hard to explore empirically for several reasons. First, it ideally requires jointly measuring the impacts of regulations on unit labor costs and the labor content of trade flows, which are rarely observed in trade datasets (Rodrik 1996). Second, cross-country differences in labor costs are correlated with productivity levels (Alesina and Perotti 1994; Mah 1997), complicating identification. A solution to overcome this challenge is to look at the effects of a domestic reform on country-level exports (Gan, Hernandez, and Ma 2016; Malgouyres and Mayer 2018). Building on this approach, I study trade in labor-intensive services across all EU countries, exploiting multiple sources of exogenous variation in tax policy and minimum wages triggered by the EU posting program.

Large trade responses to labor regulations in the context of posting are consistent with the literature showing that international capital and trade flows are sensitive to differences in corporate taxation (Griffith, Hines, and Sørensen 2010; Tørsløv, Wier, and Zucman 2018) or environmental regulations (Hanna 2010; Shapiro and Walker 2018). Those results also echo the findings that multinationals' sourcing decisions can be affected by responsible sourcing and fair trade policies (Harrison and Scorse

⁵This empirical question also relates to the earlier theoretical work that revisited the findings of classical trade theory by integrating minimum wages (Brecher 1974) or fair wages (Agell and Lundborg 1995).

2010; Boudreau 2021; Alfaro-Ureña et al. 2021; Dragusanu, Montero, and Nunn 2022). Finally, those results empirically confirm that tax reforms (fiscal devaluations) can boost exports in labor-intensive sectors (Farhi, Gopinath, and Itskhoki 2014).

From a policy perspective, this paper raises the question of whether international coordination can prevent a “race to the bottom” in social standards, particularly in the EU, where trade and migration are free but there is no coordination on tax and labor policies.⁶ While most academic debates have focused on limiting competition in corporate taxation (Avi-Yonah and Clausing 2007; Auerbach 2010), increasing evidence suggests that labor, like capital, is sensitive to international tax and regulatory differences (Kleven et al. 2020; Naidu, Nyarko, and Wang 2016).⁷ Many policies or technological changes, such as posting or remote work, could make labor more mobile and more sensitive to labor cost differentials. Furthermore, policy questions regarding factor mobility and labor cost differentials are not exclusive to international flows. Within the United States, trade in factor services also occurs and may also respond to variations in state-specific labor standards.

I. Description of the EU Posting Policy

A. A Large Trade Program for Labor Services

I study the relationship between labor market regulations and trade in a large EU program called the posting policy. Established in 1959, this law regulates temporary contracts performed locally by foreign firms. Firms located in the territory of one member state can send their workers in any other member state to perform a temporary service contract.⁸ This policy was first described by Muñoz (2024): 2 million workers (in full-time equivalents) were involved in these cross-border transactions in 2019. Posting contracts mostly cover labor-intensive activities: The first sector is construction, followed by manufacturing services, road transportation, temporary employment agencies, and agriculture. Hence, the vast majority of posted workers are blue-collar workers.

Regulatory Costs.—Posting aims to facilitate cross-border supply of services by minimizing regulatory costs. Foreign companies can directly serve customers in other EU countries; there is no need to open an establishment in that country. Foreign firms do not need to obtain a license or a work authorization to access foreign markets. They must only fill out an administrative form that allows their employees to claim formal employment and insurance in the origin country while effectively working abroad. Though the policy is intended for temporary service supply, there is no legal limit to posting contracts.

⁶Throughout the paper, I call those domestic choices “standards,” not to make the normative argument that it is better to have high taxes and minimum wages but to emphasize that some countries have a preference for higher levels of redistribution and worker protection.

⁷This question is also related to the theoretical (Lerner 1936; Feldstein and Krugman 1990; Barbiero et al. 2019; Costinot and Werning 2019) and empirical (Benzarti and Tazhitdinova 2021) work on the trade neutrality of value-added taxes. See also Kanbur and Keen (1993) and Keen and Konrad (2013) for work on international capital flows and tax competition.

⁸Posting is known as mode IV export of services in the WTO general framework for trade in services. It is not specific to the European Union and exists in other areas of the world, although mode IV is more regulated elsewhere.

Labor Market Regulations.—The general principle of posting is that firms (and their workers) are subject to regulations in the country of origin, which is defined as the country where the firm exercises its “usual activity.” Posting was initially conceived with the idea that there should be no differences in the treatment of firms exporting washing machines or construction services; in theory, all regulations should be set in the country of origin.

In the 1990s, this legal framework was changed due to concerns about unfair competition. Although trade barriers are generally not allowed in the common market, the first Posted Workers Directive was voted in 1996, with the aim to prevent “social dumping.” This new legal framework allowed member states, under certain conditions set by the European Union, to impose *some* of their domestic standards on foreign firms engaged in the program.

First, destination countries can impose some components of their labor law on exporting firms. The EU directive sets the “hard core” of domestic labor regulations that can be extended to foreign firms: maximum hours of work per week, safety and hygiene, equal pay between men and women, and minimum legal wages. The domestic minimum legal wage only applies to foreign firms that would pay their workers below that level otherwise. Only wage floors defined as “generally applicable” by domestic law bind for posted workers. Other wage floors (e.g., set at the branch or firm level) only apply to domestic workers, not posted workers. All domestic regulations that are outside the “hard core” of rules listed in the EU directive and described above do not apply to foreign firms.⁹

Second, payroll taxes become destination based if posting contracts last more than a certain duration. This duration was 12 months until 2010, 24 months from 2010–2020, and 18 months since 2020. Since 2010, payroll taxes also became destination based for temporary employment agencies located in border regions that post workers abroad.

Enforcement.—Destination countries have no right to refuse the intervention of foreign suppliers of services in their territory but can control that the posting rules are enforced. They can perform inspections on job sites where the posted workers are operating. They can also request that exporting firms fill out a form (“notification”) when operating in their territory. The enforcement of domestic standards is facilitated by the fact that exported services are performed in the importing country’s territory. This contrasts with the enforcement issues usually associated with responsible sourcing policies (Alfaro-Ureña et al. 2021), where importing countries (or multinationals) can hardly control production processes abroad. While the enforcement of “social standards” is likely much easier in posting than in other forms of trade, it plausibly remains weaker than the enforcement of domestic regulations for domestic workers. For instance, violations of some EU rules have been documented in various policy reports.

⁹Since 2020, a new EU directive voted in 2018 has expanded the range of domestic rules applicable to foreign firms. More details on the new rules are provided in Supplemental Appendix B. The empirical analysis will not cover years after 2018 that are affected by this change in regulatory framework.

B. Conceptual Framework of Trade in Services with Social Standards

I lay out a conceptual framework that clarifies how labor market regulations interact with trade in posting services. The model serves two purposes: explaining the conceptual role of changes in labor market regulations on posting flows and guiding the empirical analysis. The model suggests what regression to run, what the coefficients should identify, and outlines which endogeneity problems seem particularly relevant for the estimation.

Labor Cost Differences in Posting.—To produce services in country j , firms in origin i pay an hourly wage w_i to their employees hired in country i . Firms must pay an additional wage component to reach the minimum legal wage in the destination country $\underline{w}_j$. I call this the posting allowance. This term is zero if the destination country does not have a minimum legal wage ($\underline{w}_j = 0$), if the worker wage is already higher than the minimum wage abroad ($w_i \geq \underline{w}_j$), or if the service is produced at home ($j = i$). Firms are also subject to the payroll tax rate τ_i in their origin country, but the posting allowance is tax-exempt (Chevreux and Mathieu 2016). The hourly labor cost for services supplied by workers posted from i to j is

$$(1) \quad \left\{ \begin{array}{l} \underline{c}_{ij} \quad \text{labor cost to export from } i \text{ to } j \\ \underline{c}_{jj} \quad \text{labor cost to produce in } j \end{array} \right. = \begin{array}{l} \underline{w}_i \quad \text{wage in } i \\ \underline{w}_j \quad \text{wage in } j \end{array} + \begin{array}{l} \tau_i \cdot \underline{w}_i \quad \text{payroll taxes} \\ \tau_j \cdot \underline{w}_j \quad \text{payroll taxes} \end{array} + \begin{array}{l} \mathbf{1}_{\{w_i < \underline{w}_j\}} (\underline{w}_j - w_i) \quad \text{posting allowance for min. wage} \\ 0 \end{array}$$

Firms competing to produce services can have different wage costs because (i) they face different equilibrium wages, (ii) they have different labor tax rates and basis, and (iii) they are differentially affected by destination-specific minimum wages.

I do not model the determination of the equilibrium wage w_i . It is worth noting that in models like Balassa (1964) and Samuelson (1964), each country has the same productivity in services but wages for service workers differ across countries because of differences in outside options set by wages (productivity) in the tradable sector. In standard trade models, services are not traded. But posting allows trade in “nontradable” services, turning wage differences in those services into a potential source of cost advantage. An important question is thus whether the outside option for workers sent from Poland to France is the equilibrium wage in France or in Poland.¹⁰ The current assumption that posted workers are hired at wages w_i , while later sent abroad is similar to the modeling of immigrants’ wages in Ottaviano, Peri, and Wright (2013). In the context of posting, this seems plausible since posted workers’ wages are contracted in the home country with their usual employer.¹¹

¹⁰In Grossman and Rossi-Hansberg (2008) and Rodríguez-Clare (2010), offshoring allows firms to pay foreign workers with similar productivity at lower wages, which creates efficiency gains. In contrast, in their models, immigration does not allow to pay the same foreign workers at lower wages.

¹¹Posted workers remain employed by their employer in the origin country during the entire posting contract. Only 10 percent of workers posted to France are hired in the origin country just before being posted abroad. Therefore, the vast majority of posted workers have a usual employment relationship with the exporting firm, and their wages are not only set for the posting assignment abroad.

One Model of Trade in Services.—I use the Eaton and Kortum (2002) model to show how labor cost differences in equation (1) shape countries' market shares in services. There is a finite number of countries $i \in S$ and a continuum of services Ω_n that every country can produce. Services are produced by combining hours of labor with country i 's efficiency in producing services n denoted $z_i(n)$. Unit labor costs in sending countries are gross wages paid to workers divided by productivity. Supplying services from country i to country j also generates bilateral mobility costs m_{ij} . The unit cost for posting services from i to j is

$$(2) \quad \underbrace{C_{ij}(n)}_{\text{unit cost to export from } i \text{ to } j} = c_{ij} \cdot m_{ij} \cdot \frac{1}{z_i(n)}.$$

There is perfect competition across service suppliers.¹² As in Head, Mayer, and Ries (2009), each service is purchased from the country that offers the service at the lowest unit cost, including bilateral mobility costs. A service n is exported from i to j if and only if

$$(3) \quad C_{ij}(n) \leq \min_{i' \in S} \{C_{i'j}(n)\}.$$

The assumption of Frechet distributed productivity with $F_i(z) = \exp\{-(T_i z)^{-\theta}\}$ yields for each service n the probability that country i provides the lowest-price service in country j . The density distribution of productivity together with the condition for optimal sourcing choices sets the total number of postings from i to j . The share of services in country j performed by suppliers from country i is

$$(4) \quad \lambda_{ij} = \frac{T_i (c_{ij} \cdot m_{ij})^{-\theta}}{\sum_{k \in S} T_k (c_{kj} \cdot m_{kj})^{-\theta}} = T_i (c_{ij} \cdot m_{ij})^{-\theta} \Phi_j^{-1},$$

where $\Phi_j = [\sum_{k \in S} T_k (c_{kj} \cdot m_{kj})^{-\theta}]^{-1}$.¹³ Denoting S_j the total demand for labor services in country j , the volume of posting contracts exported by i to j is

$$(5) \quad \begin{cases} S_{ij} = S_j T_i (c_{ij})^{-\theta} m_{ij}^{-\theta} \Phi_j^{-1} \\ c_{ij} = \mathbf{1}_{\{w_i \geq \underline{w}_j\}} [w_i (1 + \tau_i)] + \mathbf{1}_{\{w_i < \underline{w}_j\}} [\underline{w}_j + \tau_i w_i]. \end{cases}$$

Equation (5) is the foundation of all empirical exercises that will be presented throughout the paper. The ultimate goal is to estimate the sensitivity of trade in services to labor costs c_{ij} . This is captured by the elasticity θ , which is the counterpart of the structural trade elasticity in similar gravity models (Head and Mayer 2014).

¹²This assumption implies pricing at marginal cost, which may not always be realistic. For instance, Thum-Thysen and Canton (2015) document the existence of markups in service sectors in the European Union, although those markups have declined over the past two decades.

¹³Note that λ_{ij} is the fraction of services that $j \in S$ purchases from $i \in S$. Allen and Arkolakis (2016) show that with Eaton and Kortum (2002) and Frechet productivity shocks, the fraction of goods purchased from a given origin is equal to the fraction of income spent on goods from a given origin.

II. Measuring Trade in Labor Services and Labor Costs

A. Datasets on Cross-Border Supply of Labor Services (S_{ijt})

A1 Posting Forms.—I use administrative social security forms (“A1”) issued for each posting contract within the European Union. This form is a mandatory document filed by exporting firms that workers must hold during their assignment abroad to prove their formal employment and insurance in the origin country. I collected all posting forms issued and received by 25 European member states from 2007 to 2018 from the European Commission (European Commission 2011, 2012; De Wispelaere and Pacolet 2014, 2015, 2016, 2017, 2018; De Wispelaere, De Smedt, and Pacolet 2019).¹⁴ Compared to standard custom data, this dataset directly measures the number of workers involved in the production of the service abroad. The dataset records posting flows for each origin and destination country each year. One limitation is that when workers are sent by their firm to more than one country in a given year, only one A1 form is issued by the sending country, and this form cannot be linked to a specific receiving country.

For more recent years, I observe the average duration of posting contracts by exporting countries. For a subsample of six exporting countries (Poland, Romania, Luxembourg, Lithuania, Czech Republic, and Hungary), I also obtained the same bilateral A1 forms flows further disaggregated by sector categories. Finally, I use the gravity dataset from Head and Mayer (1948–2018, 2014, 2021) to obtain additional information on bilateral trade in goods, GDP, and geographic distance for each origin-destination country pair in my posting dataset.¹⁵

Additional Posting Registries.—In France, Belgium, and Luxembourg, I have access to granular national datasets on posted workers that supplement the A1 forms. The datasets provide more information on the duration of posting assignments, the wages paid to posted workers, and five-digit sector activities of posting contracts.¹⁶

I have access to the IGSS (2002–2020) matched employer-employee dataset covering all firms in Luxembourg with information on whether each of their employees is posted abroad each month. I use Direction Générale du Travail (2005–2016b) data on the number of posting contracts performed in France since 2005, collected from Direction Générale du Travail (2016, 2019), as well as DGT-DARES (2016–2020) data on disaggregated posting contracts in France for 2017–2020, with information on posting contracts’ duration and posted workers’ wages. Finally, I use ONSS (2008–2019) data on the number of posting contracts performed in Belgium since 2010. All datasets are described in detail in Supplemental Appendix C.

¹⁴The countries are Austria, Belgium, Bulgaria, Czech Republic, Germany, Denmark, Estonia, Spain, Finland, France, Croatia, Hungary, Ireland, Italy, Lithuania, Luxembourg, Latvia, the Netherlands, Poland, Portugal, Romania, Sweden, Slovenia, Slovakia, and the United Kingdom.

¹⁵I also use CEPII (2007–2023) data on trade in goods by product code from the BACI database (Gaulier and Zignago 2010) and the OECD (2005–2023) dataset on trade in services by EBOPS 2010 service categories. Finally, I also collected Eurostat (2009–2018f) data on unemployment rates, Eurostat (2009–2018e) data on price level indices and Eurostat (2009–2018a) data on employment by industry.

¹⁶Those registries also register all incoming postings regardless of the number of countries the posted workers are serving in during the year.

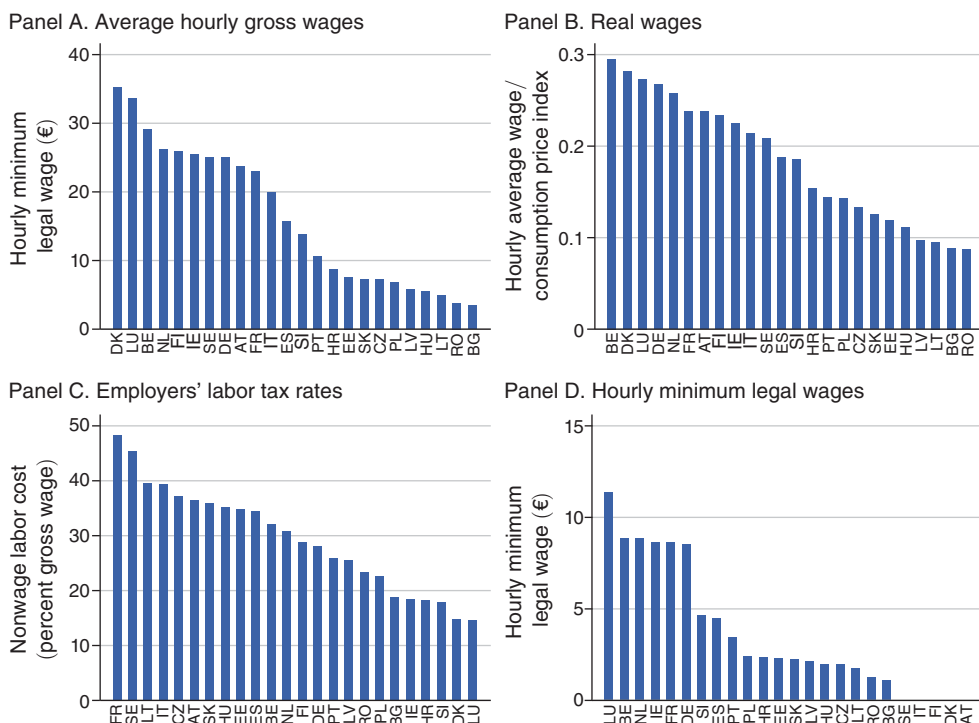


FIGURE 1. WAGES, PAYROLL TAXES, AND MINIMUM WAGES IN THE EUROPEAN UNION

Notes: This figure shows different components of hourly labor costs in Europe in 2015, using data collected from Eurostat. Panel A plots average nominal hourly gross wages paid in the country. Panel B divides gross hourly wages by a harmonized consumption price index to account for differences in prices across EU countries. Panel C plots employers' effective payroll tax rates, computed as social security contributions and other labor taxes net of subsidies, in percent of gross wages paid to employees. Panel D plots nationwide statutory minimum legal wages.

B. Datasets on Employers' Labor Cost (c_{ijt})

I measure each component of hourly labor cost for each importing and each exporting country, taking into account the specific rules in the EU posting program. I use the labor cost indicator (LCI) dataset from Eurostat (2009–2018b), built from national firm-level administrative datasets, available from 2009 to 2018. It measures the average hourly gross wage paid by employers, and the nonwage component of labor costs, which is based on paid social security contributions and labor taxes minus subsidies received by employers. To distinguish payroll taxes from other taxes and subsidies, I also collect social security contribution rates for employers from the OECD (2009–2018) Taxing Wages Dataset. This more directly reflects policy changes by recording the legal rates faced by employers in each country. Finally, I collect information on nationwide minimum legal wages from Eurostat (2009–2018c) for the same period.

Figure 1 illustrates the disparities in gross wages, payroll taxes, and minimum wages within the European Union. Despite the absence of regulatory trade and migration barriers in the single market, panel A shows large differences in gross hourly wages. Panel B adjusts those wages by prices. There is less dispersion in real than nominal wages, but the EU still experiences substantial heterogeneity in real wages.

Panels C and D show large differences in payroll taxes and minimum wages. For instance, France and Luxembourg have similar levels of wages, but payroll taxes are three times higher in France. Countries with similar wage levels do not mechanically have similar minimum wages. The minimum wage is set at 60 percent of the median national wage in France, against 45 percent for the Netherlands. Some countries, like Sweden or Germany (before 2015), do not have a minimum legal wage but branch- or firm-level wage floors instead.

Figure 1 only captures some of the heterogeneities in labor market regulations across EU member states. For instance, rules on employment termination or worker protection also vary across countries and matter for labor costs. I will not be able to measure those sources of wage cost heterogeneity nor isolate their causal effects on posting use. My main empirical design will instead control for those (permanent) differences in labor regulations by only exploiting changes in the components of wage costs that I can accurately measure.

C. Descriptives on Wages and Incidence of Labor Rules in Posting

Combining information on hourly gross wages, employers' effective labor tax rates, and minimum wages with the rules described in equation (1), I can reconstruct c_{ijt} , the hourly wage cost for employers exporting labor services from country i to country j in year t . This requires the assumption that nationwide average wages are a good proxy for wages paid to posted workers. For instance, if the wage for posted workers in the origin country is above the average wage in that country, I could wrongly infer that destination-level minimum wages are binding for exporting firms. Fortunately, micro-level data on posted and domestic workers' wages in France help gauge the credibility of this assumption.

Supplemental Appendix Figure D.4 starts by showing that most workers posted to France are paid exactly at the French minimum wage. This is evidence that the wage floor enforced by the EU policy is binding and that exporting firms would pay lower wages absent the minimum wage rule. Instead, French workers competing with posted workers are paid way above the French minimum wage. In 2018, the average wage of French workers employed at French firms importing posting services was €21. The nationwide average hourly wage was €24, and the French minimum wage was around €10. Thus, foreign and domestic firms pay different wages to hire workers for similar tasks. This wage gap could stem from differences in workers' ability. However, Muñoz (2024) finds substantial within-workplace differences in gross wages paid to posted and domestic workers, even after accounting for tenure or productivity proxies. This suggests that there are enough institutional and information frictions to enable wage discrimination between otherwise similar posted and domestic workers, as in offshoring (Grossman and Rossi-Hansberg 2008) or outsourcing (Goldschmidt and Schmieder 2017; Katz and Krueger 2019).¹⁷ Those gross wage differences (that do not account for differences in payroll taxes) also

¹⁷ Posted workers are bound to their exporting firms, making it hard to switch jobs during the contracts or to renegotiate wages. This wage gap suggests that competition between exporting firms is too weak to pull up wages of foreign workers as much as for domestic workers. Albert and Monras (2022) also emphasize that temporary migrants can have lower reservation wages than natives.

reflect differences in labor regulations. For instance, a French worker will be paid extra for working at night or on the weekends, but those rules won't apply to posted workers.

Figure 2 then shows that firms located in different countries are differentially affected by destination-specific minimum wages. Most posted workers sent to France from countries where the average wage is below the French minimum wage are paid exactly at the French minimum wage (panels A to C). But posted workers' wages are much less concentrated around the French minimum wage when they are sent from high-wage countries (panels D to F). The posting allowance that foreign firms must pay to reach the French minimum wage is thus large for firms located in low-wage countries but is zero for others. This creates a dyadic source of variation in c_{ij} , as I will discuss in Section IIIB.

III. Gravity Model of Trade in Labor Services

This section presents evidence based on all sources of variation in bilateral posting flows, wages, payroll taxes, and minimum wages in 25 member states over the period 2009–2018. First, I present cross-country correlations between labor cost differentials and posting flows. This provides suggestive evidence that labor costs matter for the geography of trade in services in the long run. Second, I present results based on a theory-consistent estimation of the gravity model for cross-border labor service trade, with inference based on a set of exporter-year and importer-year fixed effects.

A. Cross-Sectional Correlation between Trade Flows and Labor Costs

Figure 3 presents cross-sectional evidence on the relationship between bilateral trade flows and labor cost differentials. In each panel, I depict the best linear fit using a univariate regression with no country weights. I also estimate corresponding elasticities by regressing the log bilateral trade flows on the log origin-destination labor cost differential.

Panel A focuses on trade in posting services. Panel A1 shows the binned scatterplot of (log) bilateral posting flows (measured by number of posting A1 forms) and (log) origin-destination labor cost differentials for all country pairs during the 2009–2018 period. A tight relationship holds between the two, with a cross-sectional correlation of $-0.58(0.08)$. Panel A2 shows an equally large and even tighter correlation when focusing on the nonwage component of labor costs, e.g., differences in social security contributions and other labor taxes. The negative correlation holds when focusing on bilateral posting flows in values (Supplemental Appendix Figure D.8) or after controlling for average wage difference (Supplemental Appendix Figure D.11).¹⁸

Panel B of Figure 3 focuses on trade in manufacturing goods, which by definition, must be less labor intensive than posting services. While bilateral trade in goods and

¹⁸The elasticity is equally large when weighting posting flows by wages (Supplemental Appendix Figure D.7), duration (Supplemental Appendix Figure D.6), or when using employers' SSC statutory rates as a measure of labor costs (Supplemental Appendix Figure D.10). Using log odd ratios $\ln(S_{ij})/\ln(S_{jj})$ as an alternative outcome variable yields an elasticity of $-0.52(0.05)$ (Supplemental Appendix Figure D.9).

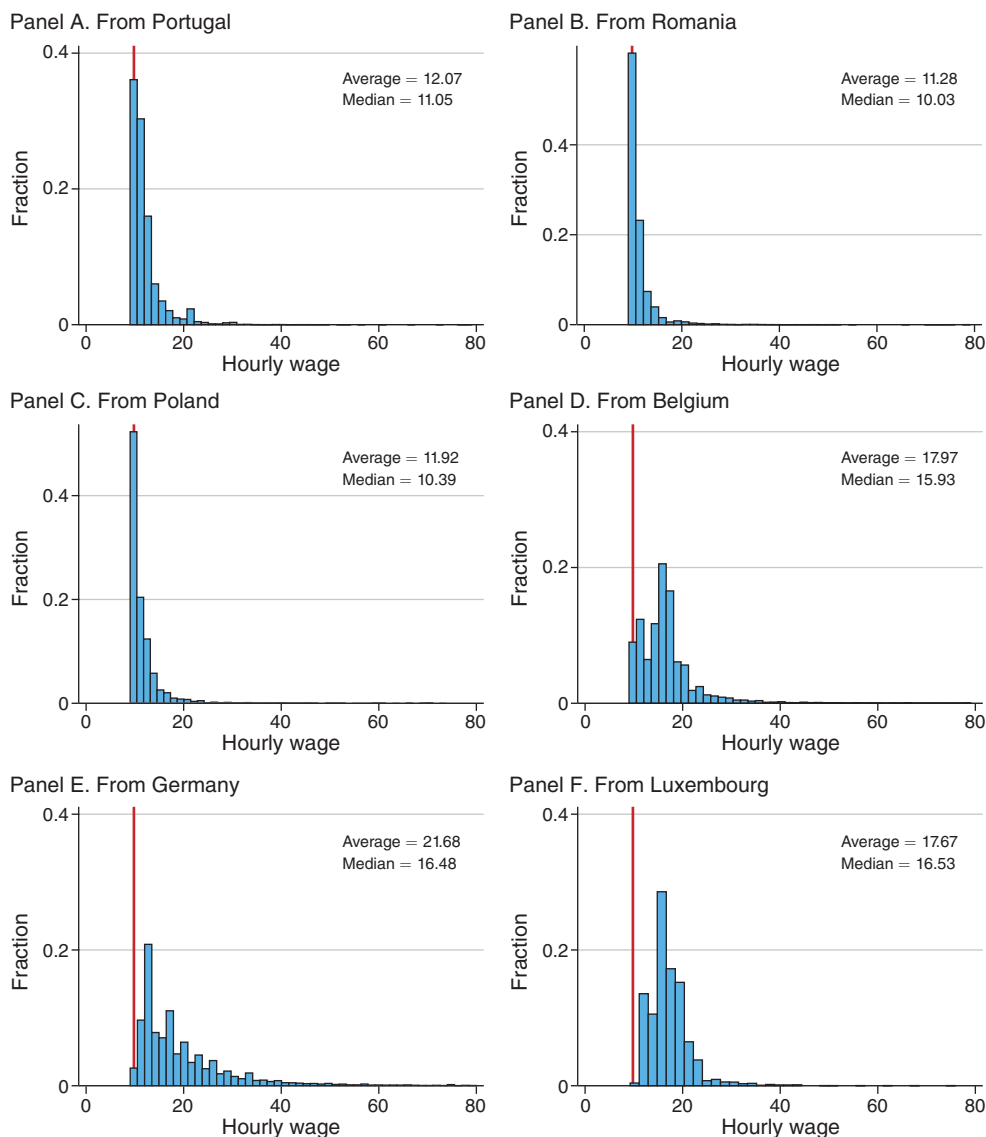
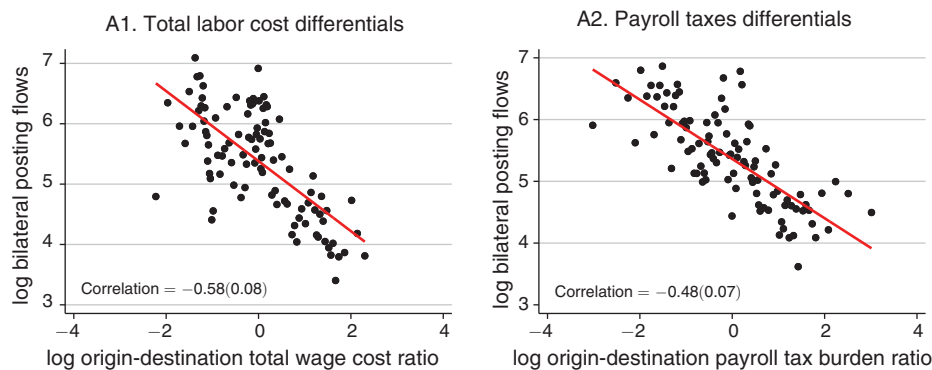


FIGURE 2. BINDINGNESS OF FRENCH MINIMUM WAGE BY ORIGIN OF EXPORTING FIRMS

Notes: This figure illustrates the degree to which the domestic minimum wage of an importing country (France) binds for exporting firms located in different origin countries. Each panel plots the distribution of hourly wages paid to workers employed by exporting firms in each of those countries and supplying posting services within French territory in 2018. Per EU law, exporting firms must comply with the French minimum legal wage during the posting contract. The French minimum legal wage that year is depicted by the vertical red line. Each figure displays the median and average nominal hourly wage paid by firms located in each of those countries when supplying services in France.

services has been shown to react similarly to gravity forces (Eaton and Kortum (2018), Figure 3 shows a fundamental difference in the way those flows react to labor costs in the European Union. Countries with similar levels of labor costs are also the ones that trade large amounts of goods, a pattern that was not observed for trade in posting services. The cross-country correlation between bilateral trade in

Panel A. Trade in posting services (labor intensive)



Panel B. Trade in all goods (less labor intensive)

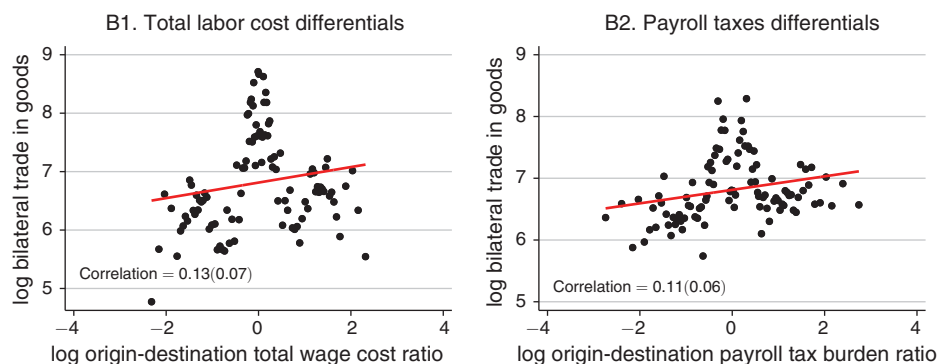


FIGURE 3. STEADY-STATE CORRELATION BETWEEN TRADE FLOWS AND LABOR COST DIFFERENTIALS

Notes: This figure depicts the relationship between the log of bilateral trade flows and the log of labor cost differentials, for a given country-pair during the 2009–2018 period. The top panel uses bilateral posting flows (physical services) measured by number of posting contracts (A1 forms) exported from one country to another. The bottom panel uses bilateral trade in goods (in values) from Head and Mayer (2014). Panels A1 and B1 focus on total hourly labor cost differentials, after accounting for the specific payroll tax rates and minimum wages that apply to posted workers in Panel A1. Panels A2 and B2 focus on differences in the payroll tax burden, measured as the amount paid for social security contributions and other labor taxes per hour of labor. The red line depicts the best linear fit using a univariate regression. The correlation between log bilateral trade flows and the log origin-destination labor cost differentials is displayed in each figure, with standard errors clustered at the country-pair level in parentheses.

goods and labor cost differentials is weak and if any, positive, with a point estimate of 0.13 (0.07). These different patterns may have several explanations. Trade in goods involves an entire bundle of inputs, including capital and more skilled labor, so that differences in average wage costs are unlikely to capture true differences in unit costs. Properly measuring the differences in unit costs driven by wage variations would require data on the labor content of trade flows, which is hard to come by in practice (Rodrik 1996).¹⁹ Therefore, focusing on services, which are labor intensive, is more likely to provide a credible estimate of the elasticity of trade flows

¹⁹ If labor is not the only input, the elasticity needs to be appropriately scaled by the share of wages in the cost function, as discussed in Eaton and Kortum (2002).

with respect to wage costs.²⁰ Furthermore, other sources of relative cost differences could also drive bilateral trade in goods relative to services, such as similar preferences (Linder 1961), or because higher wages are correlated with goods' quality.

Are the differential patterns observed in Figure 3 driven by a mechanism that is specific to services or specific to (low-skill) labor intensity? Panel A of Supplemental Appendix Figure D.13 focuses on financial services, which are also services but are not as intensive in low-skill labor. The relationship between trade in financial services and labor cost differentials is positive, with a large spike of flows around zero, exactly like trade in goods. Panel B then studies trade in footwear and apparel, which are goods that rely more intensively on low-skill labor (Romalis 2004). The relationship with labor cost differentials does not follow the general pattern observed for all trade in goods in Figure 3. There is no spike in flows around zero, and the correlation is negative, although the slope is much less pronounced than for posting services. Supplemental Appendix Figure D.13 confirms that the differential patterns showed in Figure 3 are driven by low-skill labor intensity rather than something specific to all services.

Figure 3 is informative of the steady-state impact of labor costs on trade in posting services. Many factors are simultaneously correlated with labor costs and competitiveness in services, including differences in productivity levels. Static comparisons across countries also load the effects of differences in other regulatory components of labor costs. To properly estimate the causal effect of labor costs on posting flows, it is crucial to exploit exogenous variation in labor regulations while controlling for these simultaneous factors.

B. Gravity Estimation

Identification.—My goal is to estimate the sensitivity of trade in services to labor costs. For this, I take equation (5)—the gravity prediction from the model of trade in services derived in Section IB—to the data. I estimate the model using the following specification:

$$(6) \quad \begin{cases} S_{ijt} = \exp\{-\theta \log(c_{ijt}) + \gamma_{ij} + \gamma_{it} + \gamma_{jt} + m_{ijt} + \varepsilon_{ijt}\} \\ c_{ijt} = \mathbf{1}_{\{w_{it} \geq \underline{w}_{jt}\}} [w_{it}(1 + \tau_{it})] + \mathbf{1}_{\{w_{it} < \underline{w}_{jt}\}} [\underline{w}_{jt} + \tau_{it} w_{it}]. \end{cases}$$

My main outcome variable is S_{ijt} , the number of posting contracts (posting A1 forms) exported from i to j at time t .²¹ The main parameter of interest is θ , the elasticity of posting services to labor costs. Origin-destination fixed effects (γ_{ij}) control for time-invariant posting costs (e.g., distance). I include a dummy m_{ijt} for changes in bilateral posting rules.²² Origin-year (γ_{it}) and destination-year (γ_{jt}) fixed effects, when included, absorb all i -specific and j -specific terms in equation (5) that can vary over time (for instance, productivity T_{it} in the origin country or multilateral resistance term Φ_{jt} in the destination country). Standard errors are clustered at the

²⁰This is particularly true for trade in posting services, where capital stays put but firms decide from which country to hire labor.

²¹Like in standard gravity models, equation (5) could be expressed and then estimated using trade flows in expenditures or in volume. Since my datasets measure the number of posting contracts better than the value associated to those contracts, I use volume (numbers) of postings as my main outcome.

²²The bilateral variation in posting liberalization events is summarized in Supplemental Appendix Table B.1.

destination-year level, but I show robustness to alternative inference procedures. I now describe the sources of plausibly exogenous variations in c_{ijt} that identify θ .

- (i) **Payroll Tax Rates in Origin:** Workers sent from different countries are subject to different tax rates within the same receiving country, and those taxes vary over time. Payroll tax reforms create origin-specific variation in wage costs within a given destination. For instance, a payroll tax reform in Poland lowers wage costs for Polish firms supplying services in France but not for Spanish firms operating in France the same year. I can control for a destination-year fixed effect γ_{jt} that absorbs all changes in demand and supply of services in the destination country. This is usually the best available source of variation to study how international movements of labor respond to tax reforms (Kleven et al. 2020).²³ However, this source of variation does not allow to control for an origin-year fixed effect γ_{it} . Therefore, other reforms or productivity shocks occurring simultaneously in the origin country and boosting its exports of posting services can be a threat to identification.
- (ii) **Minimum Wages in Destination:** The enforcement of destination minimum wages for some origin-destination pairs but not others introduces bilateral variation in c_{ijt} . Suppose that country j is increasing its minimum wage $\underline{w}_{jt}$ in a given year. The reform affects c_{ijt} for all countries $i \in S$ such that $w_{it} < \underline{w}_{jt}$. But any country $i' \in S$ with $w_{it} \geq \underline{w}_{jt}$ is not affected (bound) by the minimum wage reform. There is origin-specific variation in wage costs for a given destination j , as soon as $\underline{w}_{jt}$ is binding for some origin countries but not others in a given year. Furthermore, the change in $\underline{w}_{jt}$ affects country i 's wage costs when exporting to j but not when exporting from i to all other destination countries $j' \in S$. There is also destination-specific variation in wage costs for a given origin i , as soon as $\underline{w}_{jt}$ is binding for i in a given year.²⁴ Similarly, dyadic variations arise when considering introduction instead of yearly changes in minimum wages.²⁵

The heterogeneous bindingness of destination minimum wages creates nonlinearities in c_{ijt} . I can thus control for both origin-year (γ_{it}) and destination-year (γ_{jt}) fixed effects.²⁶ With this most restrictive set of fixed effects, the identification is comparing bilateral posting flows in country pairs that are affected (posting allowance is positive) or not affected (posting

²³For instance, Kleven et al. (2014) exploit *within-destination* differences in tax rates applied to foreigners and domestics. In the setting of posted workers, different tax rates apply not only to domestic and foreign workers but also to foreign workers sent from different origin countries.

²⁴Note that if country i is not bound by any destination minimum wage in any $j \in S$ in a given year (i.e., the posting allowance is zero for all destinations), then its labor cost $c_{ijt} = w_{it}(1 + \tau_{it})$ is the same for all destination countries and is subsumed in the fixed effect γ_{it} . In my dataset, between 50 percent and 70 percent of sending countries are bound by at least one destination minimum wage in a given year and therefore exhibit different wage costs c_{ijt} conditional on an origin-year fixed effect γ_{it} .

²⁵Over the period I study, only Germany introduced a legal minimum wage. This is the largest variation in minimum wage in my sample. I study this reform in detail, using a difference-in-difference strategy in Section IVA. In my dataset, there are 161 events of yearly changes in the height of the nationwide minimum legal wage; the average change is +5.3 percent.

²⁶Those nonlinearities are graphically illustrated in Supplemental Appendix Figure D.3.

allowance is zero) by changes in the legal minimum wage in a given destination.²⁷ For instance, an increase in the French minimum wage increases wage costs for Polish firms exporting to France but not for German firms exporting to France, nor for Polish firms exporting to Spain. The French reform creates bilateral variation in wage costs, even after accounting for Poland-year and France-year fixed effects. Those fixed effects absorb all shocks, in both France and Poland, occurring at the same time that the French minimum wage reform, and affecting posting flows within that country pair.

- (iii) **Payroll Tax Base:** Third, because payroll taxes are paid on the basis of origin-level wages, there is bilateral variation in c_{ijt} across country pairs bound by a given destination-level minimum wage (i.e., conditional on the posting allowance being positive). This dyadic variation can only be separated from origin-year and destination-year fixed effects using a log transformation of c_{ijt} .²⁸ I will show in Section IIIB that this source of bilateral variation is not quantitatively large in practice and does not drive the identification of θ .

Incidence: A last mechanism to consider for the identification of θ is that gross wages can in principle adjust to labor regulation reforms. For instance, a payroll tax increase might be passed through to wages, leaving net labor costs unchanged, although nominal rigidities, including minimum wages, may restrict those market-level wage adjustments.²⁹ I estimate trade responses to total wage costs after a tax change, which should partially capture potential wage responses to tax reforms. The incidence of minimum wage reforms is clearer because the posting allowance directly constrains gross wages paid to employees. However, exporting firms could pay the higher destination minimum wage during the posting contract and then reduce workers' wages upon their return, again leaving net labor costs unaffected. Muñoz (2024) shows that in practice, exporting firms do not decrease posted workers' wages after the end of the posting contract.

Interpretation of the Posting Elasticity: As discussed in Muñoz (2024), posting is only one of the four economic freedoms that facilitate trade in factors within the European Union. For instance, the free movement of workers allows foreign workers to be directly hired in the destination country. To interpret θ , it is important to explore whether changes in c_{ijt} cause a shift from posting to other forms of factor trade in the single market. To explore this question, I will estimate migration and nonposting trade responses to changes in c_{ijt} .

Results.—This section reports estimation results of the theory-consistent estimation of equation (6) in the EU-wide dataset. Column 1 of Table 1 presents the

²⁷ In spirit, this is similar to the identification strategy in Boehm, Levchenko, and Pandalai-Nayar (2023) comparing the effects of a given MFN tariff change on trade flows from countries bound or not bound by MFN tariffs.

²⁸ Without the log transformation, the bilateral wage cost conditional on the posting allowance being positive, $c_{ijt} = \underline{w}_{jt} + w_{it}\tau_{it}$, is additively separable in origin-year and destination-year specific terms. In that case, bilateral wage costs are collinear to destination-year and origin-year fixed effects.

²⁹ Similarly, targeted tax reforms (e.g., at sectors or exporters) are often too narrow to lead to overall wage adjustments (Saez, Schoefer, and Seim 2019).

TABLE 1—ELASTICITY OF TRADE IN LABOR SERVICES TO LABOR COST CHANGES

Variables	OLS (1)	OLS (2)	PPML (3)	MPPML (4)	MPPML (5)	MPPML (6)
$\log(c_{ijt})$	−0.760 (0.198)	−0.802 (0.209)	−1.167 (0.130)	−2.118 (0.386)	−1.189 (0.204)	−1.253 (0.225)
Observations	5,539	5,291	5,532	5,532	5,532	5,767
R^2	0.959	0.930				
Pair fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Destination $\times$ Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Origin $\times$ Year fixed effects	No	No	No	No	Yes	Yes
Internal flows	Yes	No	No	No	No	Yes

Notes: This table summarizes coefficients estimated from equation (6) on the full matrix of bilateral posting flows within the EU merged with data on employers' labor cost for each country pair for 2009–2018. The dependent variable is the number of postings from i to j at time t (S_{ijt}), in log for specifications relying on a log-linear version of equation (6) (OLS estimation), in levels for specifications relying on the multiplicative form of equation (6) following Silva and Tenreyro (2006) ("PPML" Poisson pseudo maximum likelihood estimator), and in shares λ_{ijt} following Eaton, Kortum, and Sotelo (2012) ("MPPML" Poisson pseudo maximum likelihood estimator with trade shares as dependent variable). All specifications include destination-origin fixed effects, a dummy equal to one if posting regulations between country i and j change at time t , and a destination-year fixed effect to control for the multilateral resistance term (Φ_{jt}). Robust standard errors clustered at destination-year level are displayed in parentheses.

OLS specification after taking logs of equation (6). It includes pair fixed effects and destination-year fixed effects, which absorb destination-specific shocks correlated with demand for posting services in country j . In the model, consumers choose among and consume services supplied by domestic and foreign firms. Thus, I also include services supplied domestically (S_{jji}) in the estimation sample.³⁰ The estimated elasticity is $-0.75(0.2)$, which is a bit larger than the steady-state correlation in Figure 3. Column 2 repeats the same specification but removing internal flows from the estimation. The estimate remains stable, with a coefficient of $-0.80(0.2)$.

Column 3 repeats the specification from column 2 but uses a Poisson pseudo maximum likelihood (PPML) estimator. This accounts for the fact that with OLS, the log of the multiplicative gravity equation can be biased by zero flows and heteroskedastic error terms (Silva and Tenreyro 2006). The estimated elasticity is $-1.2(0.1)$.

Next, I estimate a Poisson regression using trade shares λ_{ijt} instead of levels S_{ijt} as my outcome variable, following Eaton, Kortum, and Sotelo (2012). This estimator avoids biasing estimates of the structural parameters toward large countries and is validated by Monte Carlo simulations showed in Head and Mayer (2014). The implied trade elasticity displayed in column 4 is now larger, with an estimate of $-2.1(0.4)$.

One remaining source of endogeneity lies in unobserved shocks in the origin country. For instance, a payroll tax reform could be implemented jointly with other policies that boost exporting firms' productivity in services, leading the estimates to be biased upward. The specification in column 5 of Table 1 introduces both origin-year and destination-year fixed effects. This controls for all unobserved shocks in origin and destination countries, including changes in wage levels in both countries. The estimated coefficient becomes smaller than the coefficient in column 4 but remains large and statistically significant, with a point estimate of $-1.2(0.2)$.

³⁰ S_{jji} is computed as the number of domestic workers in j in the sectors engaged in posting minus the number of workers posted from j to other countries.

Adding internal flows to this specification in column 6 yields a similar estimate of $-1.3(0.22)$.

Robustness: I perform additional tests to gauge the robustness of the results. Column 1 of Supplemental Appendix Table D.2 shows the preferred specification when clustering standard errors at the origin-destination level, while column 2 accounts for two-way clustering at the origin-year and destination-year level. The elasticity remains statistically significant at the 1 percent level. Column 3 investigates potential heterogeneities by excluding postings outside the eurozone. Column 4 shows unchanged estimates after I account for posting contracts with multiple destination countries.

I also investigate how my estimates of θ vary when varying the measurement of average wages applicable to posted workers. Supplemental Appendix Table D.3 shows that the posting elasticity is slightly smaller when using average wages in sectors that engage in trade in posted services: It is $-1.0(0.18)$ against $-1.2(0.2)$ in the baseline. Using the national average wage in the construction sector to build c_{ijt} , the elasticity displayed in column 5 of Supplemental Appendix Table D.4 is $-0.87(0.16)$. I also build an alternative measure of c_{ijt} where the posting allowance is taxed at rate τ_{it} , meaning that $c_{ijt} = \underline{w}_{jt}(1 + \tau_{it})$ if $w_{it} \leq \underline{w}_{jt}$. The elasticities, presented in Supplemental Appendix Table D.5, are very similar to the estimates presented in Table 1. Hence, differences in payroll tax basis do not appear to be the main drivers of the identification of θ .

It is also possible that firms located in different countries must pay different compensating differentials, on top of the posting allowance, when sending workers to a given location, which would lead to some measurement error in c_{ijt} . Panel A of Supplemental Appendix Figure D.5 shows no positive relationship between the average wage paid to workers sent from country i to France and the distance between i and France: Firms do not pay higher wages to posted workers traveling higher distance.³¹ I also observe nonmonetary benefits, which are rarely measured in datasets on migrant workers. Panels B, C, and D confirm the existence of in-kind compensating differentials: The share of posting contracts with employer-provided housing, food, and transportation is greater when workers are sent from more distant countries. When time-invariant, these compensating differentials are captured by the fixed effect γ_{ij} . If they vary over time, compensating differentials are potentially correlated with ε_{ijt} . For instance, posted workers could demand compensation if there is uncertainty linked to the short duration of jobs abroad, and the value of this compensation would be affected by the degree of labor market slackness at home. Supplemental Appendix Figure D.12 shows that differences in labor market conditions correlate with trade in posting services, but the correlation is nil after controlling for wage differentials.³²

³¹To be more specific, Polish workers sent to France are paid more than Polish workers staying in Poland because of the posting allowance (Polish posted workers receive the French minimum wage, while Polish workers at home are paid the Polish wage). What Supplemental Appendix Figure D.5 shows is that Polish workers posted to France do not receive higher wages than Spanish workers posted to France, even if Poland is more distant from France than Spain. Ideally, this analysis would control for origin-year and destination-year fixed effects, as in Bryan and Morten (2019), but I only observe wages of workers posted to one destination country, France.

³²One possibility for a weak demand for insurance against short-term jobs is that posted workers have a usual employment with the exporting firm and are employed at that firm at home (Muñoz 2024).

Effects on Other Margins of Trade in Factors: The estimation structure of the gravity model allows me to test for other margins of responses when labor costs of posted workers increase. In Supplemental Appendix Table D.6, I replicate Table 1 using bilateral trade in goods as the outcome.³³ An increase in c_{ijt} slightly increases bilateral trade in goods, contrasting with the negative effect on posting flows. This indicates that the baseline estimates do not capture posting responses to unobserved shocks that would affect exports of less labor-intensive activities too. Instead, the positive sign suggests a minor substitution effect between exports of posted workers and exports of goods in response to stricter posting-specific labor regulations. However, the estimate in column 4 is too noisy to rule out a null effect.

A more natural substitute for trade in labor services is standard immigration. In Supplemental Appendix Table D.7, I repeat my analysis using the number of immigrants from country i in country j at time t as the outcome.³⁴ Across all specifications, I find small positive elasticities, suggesting that firms hire more foreign workers under domestic contracts when the cost to import labor services increases. In column 4, I control for origin-year and destination-year fixed effects, as push and pull factors (such as wage changes in the origin or destination) affect standard migration through channels other than their effects on posted workers' costs. After only exploiting bilateral changes in wage costs caused by posting-specific rules, the migration effect is not statistically different from zero anymore.³⁵

Summing-Up: The estimation of the gravity model shows that trade in services is sensitive to labor costs: A 1 percent increase in c_{iji} decreases postings from i to j by 1.2 percent. There is some substitution from posting to other trade margins, although the effects are not statistically different from 0: Bilateral trade in goods increases by 0.07 percent, and bilateral migration by 0.2 percent, in response to a 1 percent increase in posting-specific labor costs. If the gravity model, in which identification and estimation is microfounded on a standard model of trade in services, gives a baseline of the elasticity, it has two limitations. First, it cannot directly test for the exogeneity of labor cost changes. Second, it captures year-to-year responses to changes in wage costs, but it may take several years for posting flows to fully adjust to permanent changes in payroll taxes and minimum wages.

IV. Country Case Studies

This section studies posting responses to large and exogenous sectoral reforms in labor market regulations. I study two reforms: (i) the introduction of

³³ Only firms exporting posting services must meet the minimum legal wage in the importing country; manufacturing exporters are not affected by this wage rule. Origin-specific payroll tax reforms affect both exporters of posting services and goods, but firms exporting labor-intensive services are more exposed if labor costs constitute a larger portion of their total unit cost.

³⁴ This measure is collected from Eurostat (2009–2018d) and does not include posted workers.

³⁵ While Supplemental Appendix Table D.7 provides novel evidence on how labor costs and regulations may influence immigration, the concepts used to measure posting and migration at the EU level are not sufficiently comparable to precisely estimate the substitution rate between labor imported through posting versus direct migration (see Supplemental Appendix D.5). Using French data, Muñoz (2024) estimates that a 1 percent increase in the employment share of posted workers decreases the employment share of EU and non-EU foreign workers by 0.7 percent, though the effects are driven by a substitution between posting and the existing stock of foreign-born workers, not new migration flows.

TABLE 2—REDUCED-FORM ESTIMATES OF POSTING RESPONSES TO PAYROLL TAX AND MINIMUM WAGE REFORMS

Description	Elasticity w.r.t	
	labor tax rate (1)	labor cost (θ) (2)
<i>Panel A. Sectoral labor cost reforms</i>		
A minimum legal wage for foreign firms supplying posting services in Germany was introduced in the meat processing industry in August 2014, followed by a nationwide minimum legal wage for all sectors in 2015. Foreign firms supplying services in the construction sector were always subject to a German sectoral wage floor. DD estimation comparing exports of posting services to Germany in manufacturing versus construction sectors. Placebo estimation for Germany's neighboring countries.		
	A1. German minimum wage reform (2014–2015)	
Exports	-	-1.308 (0.395)
Placebo destination	-	0.438 (0.432)
A new EU regulation required that Luxembourgian firms pay payroll taxes in the destination country when posting temporary employment agency workers abroad after May 2010. DD estimation comparing services supplied abroad versus at home for firms located in Luxembourg in the temporary employment agency sector. Placebo estimation for a sector sheltered from the reform.		
	A2. Destination-based payroll taxes (2010)	
Exports	-1.547 (0.245)	-8.518 (1.397)
Placebo sector	-0.003 (0.028)	-0.019 (0.157)
<i>Panel B. Country-level reform</i>		
Employers' social security contributions rate on all employees hired in Belgium was decreased from 33 percent to 25 percent starting at the beginning of 2016. DD estimation comparing imports (and exports) of posting services in Belgium versus France.		
	B1. Belgian tax shift (2016)	
Imports	1.388 (0.343)	3.632 (0.726)
Exports	-2.834 (1.28)	-7.179 (2.16)

Notes: This table summarizes the estimates of the services trade elasticity with respect to payroll cost of posted workers, using different quasi-natural experiments detailed in the text. Column 1 displays estimates of the elasticity with respect to the payroll tax rate, while column 2 displays estimates of the elasticity with respect to the total labor cost.

destination-based minimum wages for firms supplying services in manufacturing (Germany) and (ii) the introduction of destination-based payroll taxes for firms supplying temporary employment agency services (Luxembourg). For each country-specific case study, Table 2 presents elasticity estimates using a difference-in-differences comparison of the treatment and the control sector, before and after the reform. Those estimates capture medium-term responses, as I compare outcomes a few years before the reform to a few years after the reform.

A. The German Minimum Wage Reform

The Reform.—I study the case of Germany, where there was no national minimum wage until 2015. Prior to this, only the construction sector had a sectoral minimum wage established since 1997, which was binding for both domestic and posted workers.³⁶ Other sectors had wage floors set at the firm or branch level, which were not “generally applicable” legally and thus did not extend to posted workers. For

³⁶The Posted Workers Directive was adopted in Germany in 1997 (“AEntG” or posted worker act). This law created the first legal basis for minimum sectoral legal wage in Germany. Since 1997, sectoral minimum legal wages in Germany must be implemented through the addition of branches in the AEntG scope.

example, Romanian firms could pay their workers €2 per hour—the minimum wage in Romania—while providing nonconstruction services in Germany. This was particularly salient in the meat-processing industry, where posted workers earned wages around €3 per hour, compared to €16.2 for their German counterparts (Doelfs 2012; Erol and Schulten 2021).³⁷ By 2015, posted workers constituted half of the workforce in this sector (Wagner and Hassel 2016; Wagner 2018). The situation created tensions within the EU. Belgium filed a complaint with the EU Court of Justice in 2013, arguing that Germany gained an unfair competitive advantage in food processing by importing cheaper labor from low-wage countries. This trade dispute contributed to the introduction of a national minimum wage in Germany.

In August 2014, following pressures from the German government and the European Union, the meat-processing industry implemented a generally binding minimum wage of €7.75 per hour. In January 2015, a national minimum wage of €8.5 per hour was introduced and applied to all sectors in Germany. At the time, about 15 percent of German employees earned less than that amount (Dustmann et al. 2022). One important consequence was that the minimum wage became binding for foreign firms previously paying their workers less than €8.5 per hour. Foreign firms operating in construction were not directly affected, as they were already subject to a higher sector-specific minimum wage. The changes in wage costs for posted workers in the exposed sectors were large. Dustmann et al. (2022) document an average 6 percent wage increase for German workers previously earning less than €8.5 per hour, and a 10 percent increase for the most affected workers. In comparison, the same reform automatically raised wages for posted workers in the exposed sector by an average of 40 percent and doubled wages for the most affected workers.³⁸

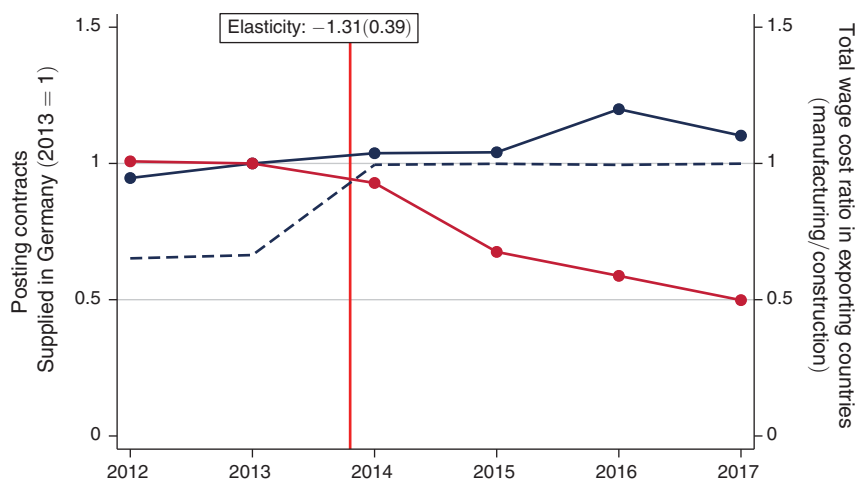
Effects of the Reform on Posting.—I study how exports of posting services to Germany responded to this reform by using data on sectoral posting A1 forms that are only available for a subset of six exporting countries: Poland, Luxembourg, Hungary, Czech Republic, Lithuania, and Romania. I observe postings exported by firms located in those countries in both the manufacturing and construction sectors, to each destination country.

Panel A of Figure 4 shows the difference-in-differences setting provided by the reform. It plots S_{jts} the yearly total number of posting contracts exported to Germany by exposed origin countries (e.g., countries with $w_{i,2013} < 8.5$), in the construction sector (control) and manufacturing industries (treated), before and after the reform depicted by the vertical red line. The series are normalized to one in 2013, the pre-reform year. I also show the sharp variation in the average exporting firms' wage costs c_{jts} in the treated sector (relative to the control sector) induced by the reform (dashed line, right y-axis). Postings to Germany in construction and manufacturing were similar in 2012, but the two series started to diverge in 2014 when the

³⁷Wages of posted workers could be as low as €1.50 per hour (Hassel, Steen Knudsen, and Wagner 2016). Wagner (2018) provides an in-depth historical perspective on the German meat-processing industry and the role of exploitative labor conditions for posted workers in its economic development.

³⁸This assumes that posted workers were paid origin-specific average wages rather than minimum wages before 2014. I do not have access to the actual wages paid to posted workers in Germany, only their numbers.

Panel A. Exports of posting services to Germany dropped in the treated sector



Panel B. No differential evolution in neighboring countries

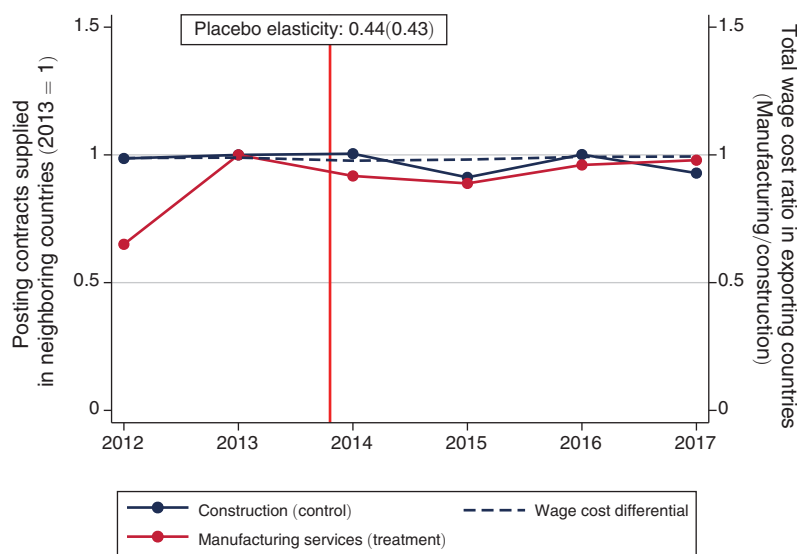


FIGURE 4. EFFECT OF IMPOSING GERMAN MINIMUM WAGE ON EXPORTING FIRMS

Notes: This figure shows posting responses to a minimum wage reform in Germany. Before 2014, only foreign firms operating in the construction sector were subject to the German (sectoral) minimum wage. Other foreign firms became liable to the German minimum wage in August 2014 (when operating in meat processing industry) or 2015 (in all other manufacturing services). The dashed line (y-axis) shows the evolution of labor costs for firms supplying manufacturing services (treatment) versus construction services (control). The top panel shows the evolution of postings in manufacturing services (treatment, red series) and construction services (control, blue series) to Germany, before and after foreign firms become liable to the German minimum wage in the treated sector. The bottom panel shows the same differential evolution in Germany's neighboring countries (France, Belgium, Austria, Netherlands, Luxembourg, Switzerland, and Denmark) where minimum wage requirements stayed constant during the period. All series are normalized to one in the pre-reform year 2013. The elasticity reported in the figure is computed with respect to the change in the total labor cost.

minimum legal wage was implemented in the meat industry. Postings to the German manufacturing industries decreased further in 2015 when all manufacturing industries became treated by the minimum wage reform. At the same time, postings in the construction sector stayed stable and close to their pre-reform level.

To get an estimate that is comparable to the baseline elasticity from the gravity estimation, I compute the corresponding elasticity from a 2SLS regression of the form

$$(7) \quad \log S_{jts} = \alpha - \theta \log(c_{jts}) + \gamma_t + \gamma_s + u_{jts}.$$

I call T_i the set of origin countries i exposed to the policy due to their low pre-reform wages, e.g., with $w_{i,2013} < 8.5$. I measure $S_{jts} = \sum_{i \in T_i} S_{ijts}$, the number of postings (A1 forms) from exposed origin countries $i \in T_i$ to destination j , Germany, in year t and sector s . I take the log of S_{jts} as my outcome variable. There are two sectors in the regression: manufacturing services (treated) and construction (control). To exploit exogenous variation in labor costs from the policy change, I use the reform interaction $\mathbf{1}_{\{s=\text{manuf. services}\}} \times \mathbf{1}_{\{t \geq 2014\}}$ as an instrument for $\log(c_{jts})$.³⁹

The year fixed effects γ_t in equation (7) control for demand and supply shocks in Germany that affect the treated and control sector similarly. The identification assumption is thus that within Germany, postings in construction and manufacturing should be exposed to similar shocks and would have followed similar trends in the absence of the reform. The elasticity θ estimated from equation (7) is $-1.3(0.4)$. It is very close to the most comparable estimate in column 5 of Table 1, which also exploits heterogeneity in bindingness of minimum wage reforms but at the origin-destination-year level.

Although the absence of pre-reform trends in panel A of Figure 4 suggests that postings in the construction and manufacturing sectors are affected by similar time-varying factors, any sector-year-specific shock that would be simultaneous to the reform and would cause postings to evolve differently in that sector is a threat to identification. For instance, these findings could simply reflect that low-wage countries are getting better at supplying manufacturing services due to simultaneous productivity shocks or policies. Panel B plots the same differential evolution in Germany's high-wage neighboring countries, where minimum wage requirements for posted workers stayed constant. Posting responses in neighboring countries can be interpreted as "placebo" effects, assuming that the German reform did not have first-order impacts on postings from exposed origin countries to other destinations. Those countries kept importing construction and manufacturing posting services at the same rate after 2014. The placebo elasticity is nonstatistically different from zero and is statistically different from the elasticity estimated in Germany. This rules out sectoral shocks in origin countries that would affect their exports to other destination countries too.⁴⁰

³⁹Where c_{jts} is the average labor cost for workers posted from exposed countries to country j , $c_{jts} = \sum_{i \in T_i} c_{ijts} \times (S_{ijts}/S_{jts})$.

⁴⁰Supplemental Appendix Figure I.27 shows the same analysis but excluding exports from Romania because nationwide restrictions to postings and migration were lifted in 2014 for Romanian citizens (see Supplemental Appendix Table B.1). The graphical patterns and estimated reduced-form elasticity are similar.

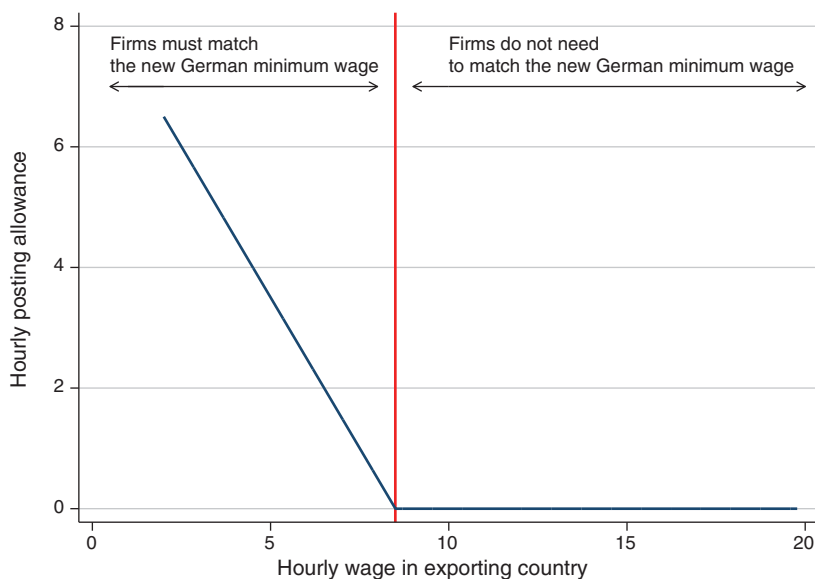
Another concern is that the minimum wage reform affected the demand for posted workers in the German manufacturing sectors through other channels than the change in wage costs for posted workers. To make progress on this issue, I take advantage of exogenous variation *within* the treated industry across sending countries. The variation comes from the *kinked* relationship between the wage cost shock caused by the reform and the pre-reform wage level in origin countries, illustrated in panel A of Figure 5 and discussed in Section IIIB. The new minimum wage only affects origin countries i with $w_{i,2013} < 8.5$. I can compare how *within* the treated German sector, posting flows from exposed versus nonexposed origins evolved differentially in the treated relative to the control sector, before and after the reform. This filters out sectoral-level effects of the reform that affects postings from all origins to the German manufacturing sector similarly. In spirit, this is similar to comparing the employment responses of domestic workers earning more or less than the new minimum wage before the reform (Dustmann et al. 2022). I estimate separately for each origin country i an equation of the form

$$(8) \quad \log S_{ijst} = \alpha + \gamma_i \cdot \mathbf{1}_{\{s=\text{manuf. services}\}} \times \mathbf{1}_{\{t \geq 2014\}} \\ + \mathbf{1}_{\{s=\text{manuf. services}\}} + \mathbf{1}_{\{t \geq 2014\}} + u_{ijst}.$$

Where S_{ijst} is the number of posting contracts (posting A1 forms) sent from country i to destination j , Germany, in year t and sector s . The interaction coefficient γ_i estimated in each regression captures the effect of the minimum wage reform on country i 's exports of labor services to Germany in the treated sector, relative to the control sector and relative to the pre-reform period. Panel B of Figure 5 shows the estimates of γ_i on the y-axis and the origin-specific change in wage cost caused by the reform in the treated sector on the x-axis. Posting responses to the minimum wage reform increase monotonically with the degree to which the minimum wage binds, suggesting that the heterogeneity captures the differential exposure to the new regulation. Exporting countries that faced the largest changes in wage costs due to their low initial level of wages at home (countries on the left of the x-axis in Figure 5) had greater reductions in their exports to Germany in the treated sector relative to the control sector and relative to the pre-reform period. In contrast, the treatment effect for the exporting country where the new German minimum wage was not binding (Luxembourg), plotted in light blue, is not statistically different from zero.

As a final falsification test, I repeat the same heterogeneity analysis focusing on destination countries that did not change minimum wage requirements during the same period. I reestimate equation (8) separately for each origin country i and each destination $j \in \{\text{Austria, Netherlands, France}\}$. The placebo responses γ_i estimated using the same specification for the same exporting countries but in Germany's high-wage neighboring countries are plotted in Figure 6. Those responses are not statistically different from zero and are not decreasing by origin-specific wages. This again rules out sector-specific shocks in exporting countries that would affect their supply of services in all destination countries, not only Germany. This also confirms that Figure 5 captures the effects of the

Panel A. Firms from low-wage countries faced larger additional costs



Panel B. Posting responses by bindingness of the reform

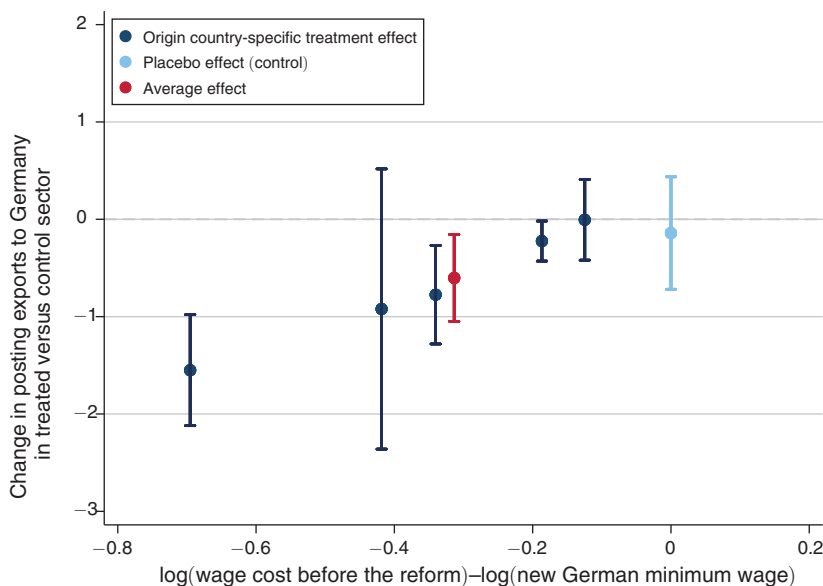


FIGURE 5. HETEROGENEOUS EFFECTS BY BINDINGNESS OF GERMAN NEW MINIMUM WAGE

Notes: Panel A shows the kinked relationship between the average wage in the exporting country and the additional wage cost that foreign firms must pay to supply services in Germany in the treated sector after the 2014–2015 reform. Panel B shows the treatment effect (and 95 percent confidence intervals) of the German minimum wage reform on exports of posting services to Germany in the treated sector, relative to the control sector, estimated separately for each exporting country. The coefficient is estimated from equation (8) and captures the differentiation evolution of postings to Germany in the treated versus control sector, after the reform relative to the pre-reform period. The blue coefficient is the estimated treatment effect for the exporting country with zero direct exposure to the reform, Luxembourg (because its minimum wage was above the new German minimum wage). The pink coefficient is the average treatment effect estimated for all exporting countries.

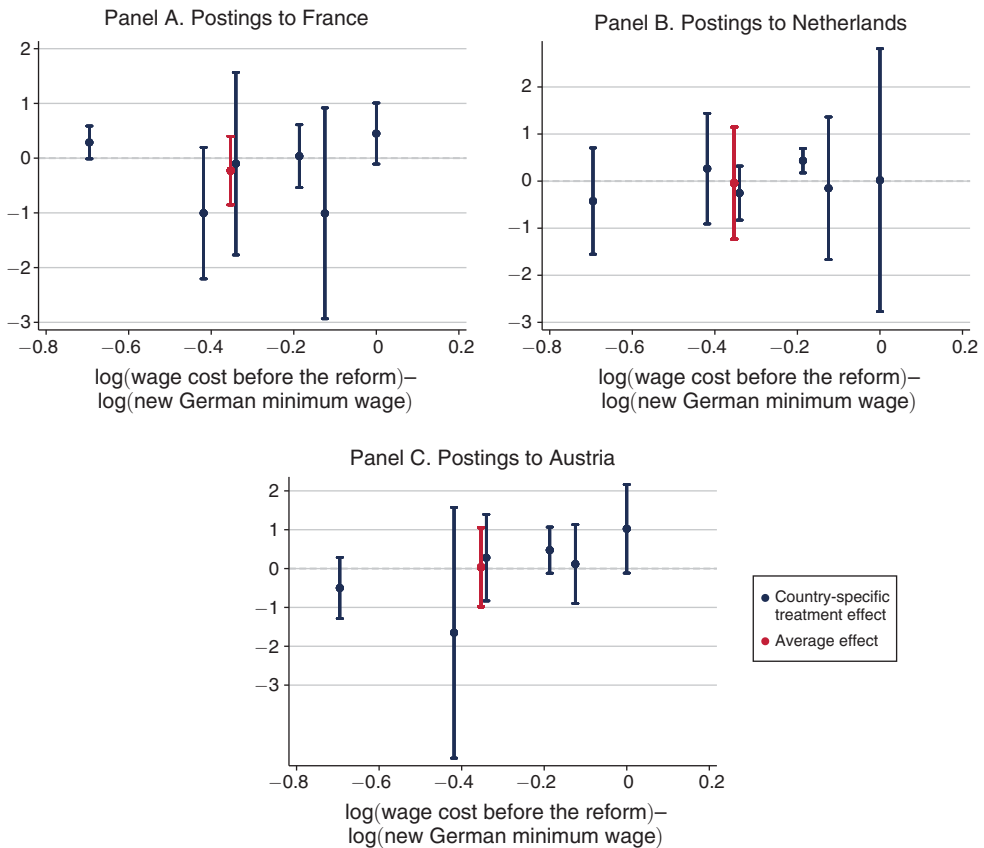


FIGURE 6. PLACEBO ESTIMATES OF THE GERMAN REFORM

Notes: This figure repeats the analysis of posting responses to the German minimum wage reform in the treated versus control sectors, showed in panel B of Figure 5, but in high-wage neighboring destination countries that did not change minimum wage requirements for posted workers in 2014–2015.

German reform, rather than a mechanical pattern that would be observed even without it.

Interpretation.—I have shown evidence that after foreign firms became subject to the German minimum wage, their exports to Germany decreased by more than half. This confirms that labor regulations, and thus labor costs, matter for international competition in labor services. It is worth emphasizing that Dustmann et al. (2022) find limited effects of the same reform on German employment (which excludes posted workers) but with substantial reallocation from less to more productive German firms. That the reform decreased foreign firms' competitiveness in supplying services through posting does not automatically imply that their workers did not reallocate to more productive firms (at home or in Germany). For instance, Mense-Petermann, Welskopp, and Zaharieva (2022) argue that the main German meat producer began employing some workers, who had previously been posted, under German employment contracts after the 2014 reform.

I test this hypothesis in Supplemental Appendix I.2, using IAB (2012–2018) dataset on German employment and wages in the processing of food and construction of buildings.⁴¹ The data cover immigrants employed under German employment contracts. The results in Supplemental Appendix Figure I.29 do not show clear evidence of increased foreign employment in food-processing jobs, suggesting that further investigation with more granular data may be needed to draw stronger conclusions.⁴² However, after 2014, the structure of economic activity in Germany's food-processing sector shifted partially, and the share of workers hired through German temporary employment agencies increased (Supplemental Appendix Figure I.30). Thus, the reform led German firms to reduce their use of cheaper labor services supplied by eastern European firms and to rely more on domestic outsourcing. Despite this adjustment, the reform appears to have permanently impacted the cost competitiveness of German meat exports. After German producers lost their ability to subcontract posted workers paid below the German minimum wage, their market shares in meat products decreased, while those of eastern and southern European countries increased (Supplemental Appendix Figure I.31). This supports the assumption that part of German meat producers' trade competitiveness was bolstered by their ability to bypass domestic minimum wage regulations through the use of imported posting services.

B. *The EU Regulation for Destination-Based Payroll Taxation*

The Reform.—Next, I study a similar reform that shifted payroll taxes—the second key component of labor regulations affecting posting wage costs—from the origin to the destination countries in a specific sector. This new EU regulation was implemented in May 2010. It required that firms located in border regions (regions transfrontalières) pay payroll taxes in the destination country when supplying their services abroad. More specifically, those firms must pay destination payroll taxes if they (i) post a worker who has not been affiliated to the home country social security system for at least a month or (ii) post a worker to its own country of residence. Such practices were mostly documented between France and Luxembourg before the reform and occurred almost exclusively through temporary employment agencies (Belkacem and Pigeron-Piroth 2016).⁴³ Furthermore, due to the difficulty in determining in which country international drivers work, a 10-year exemption was granted to firms operating in the road transportation sector.⁴⁴ The reform introduced an exceptionally narrow but sharp increase in labor taxes for some exporters but not others, keeping other determinants of foreign and domestic supply of services constant. I study the effects of this reform using an exhaustive administrative dataset

⁴¹ The IAB occupational panel only has information at the three-digit occupation level (Grienberger et al. 2023), which does not isolate meat processing from other occupations in food processing.

⁴² To get more precise estimates and establish more conclusive evidence, future analysis could use more detailed data to focus on *inflows* of EU-origin workers, include more pre-reform data, and study the meat-processing sector rather than the broader food-processing sector.

⁴³ Data from the ULEDI show that 82 percent of temporary employment agency workers in Luxembourg were French residents; this number is 20 percent when focusing on total private employment in Luxembourg. Most French workers posted to France are employed by temporary employment agencies located in Luxembourg.

⁴⁴ Those exemptions ended in April 2021 and sparked a wave of protests among workers' and employers' organizations in the sector (see Luxembourgian Parliament questions 2021n3932/3966).

from Luxembourg. I observe workers employed in Luxembourg, with information on whether they are posted or not in a given month and on the five-digit sector in which they operate.

Effects of the Reform on Trade in Posting Services.—Figure 7 shows graphically the differences-in-difference setting provided by the reform. The top panel plots the yearly number of temporary employment agency jobs in Luxembourg from 2006 to 2017 (normalized to one in 2009 just before the reform) performed abroad through posting (treatment) and domestically (control).⁴⁵ I also show the sharp variation in statutory payroll tax rates induced by the reform (dashed line, right y-axis). Payroll taxes for temporary employment agencies increased from 15 to 44 percent for posted workers (due to destination-level taxes being higher) but stayed constant for contracts performed in Luxembourg. The two series followed parallel trends before the reform, suggesting that services supplied domestically provide a credible counterfactual for exports of those services from Luxembourg. The series start to diverge immediately after 2010. The number of postings decreased by 40 percent compared to the pre-reform level. In comparison, the number of temporary employment contracts performed domestically increased during the same period. Supplemental Appendix Figure F.20 shows the same series at the monthly frequency. The drop in postings occurred in the exact month following the new EU regulation, while both series were following parallel trends in months preceding it. The high-frequency data also confirm the lack of anticipation of the reform, which contrasts with usual findings on more standard trade barriers (e.g., tariffs) where firms are able to re-time some of their exports (Fajgelbaum et al. 2020).

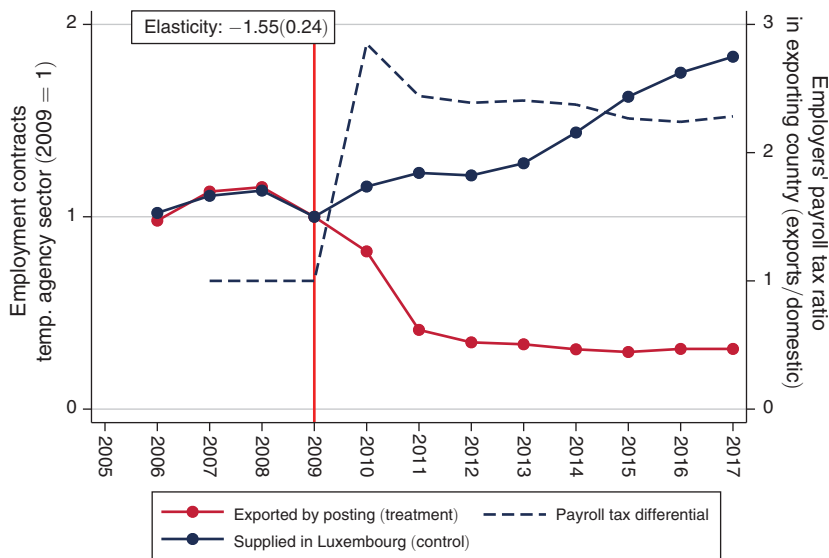
Like before, I can compute the corresponding elasticity from a gravity-based equation:

$$(9) \quad \log S_{ijts} = \alpha - \theta \log(c_{ijts}) + \gamma_t + \gamma_j + u_{ijts},$$

where S_{ijts} is total number of employment contracts in $i = \text{Luxembourg}$, operated in destination $j \in \{\text{Luxembourg}, \text{Abroad}\}$ in year t and sector s . I start by focusing on the treated sector only, hence, $s = \text{temp. agencies}$. To leverage the exogenous shock from the reform, I instrument $\log(c_{ijts})$ with the two-way reform interaction $\mathbf{1}_{\{j=\text{abroad}\}} \times \mathbf{1}_{\{t \geq 2010\}}$. The year fixed effect γ_t filters out all unobserved shocks in Luxembourg that affect similarly the supply of temporary employment services at home and abroad. The identifying assumption is thus that postings from Luxembourg in the temporary employment agency sector would have followed the same trend as the domestic supply of the same services absent the reform. The reduced-form elasticity is large and significant, with a point estimate of $-5.3(1.9)$ 3 years after the reform (a similar time window as in the German reform), and an elasticity of $-8.5(1.39)$ 8 years after the reform. Using the same procedure, I also compute the elasticity of posting flows with respect to the (log) of statutory payroll tax rate $\log(\tau_{ijts})$. The coefficient, reported in the figure and in column 1 of Table 2, is $-1.5(0.2)$.

⁴⁵ The dataset gives the number of active unique jobs in a given month. A “job” or employment contract is defined as the unique combination of a worker, an employer, and a start and end date; see Supplemental Appendix C.2.

Panel A. Exports of posting services dropped in the exposed sector



Panel B. No differential evolution in the sheltered sector

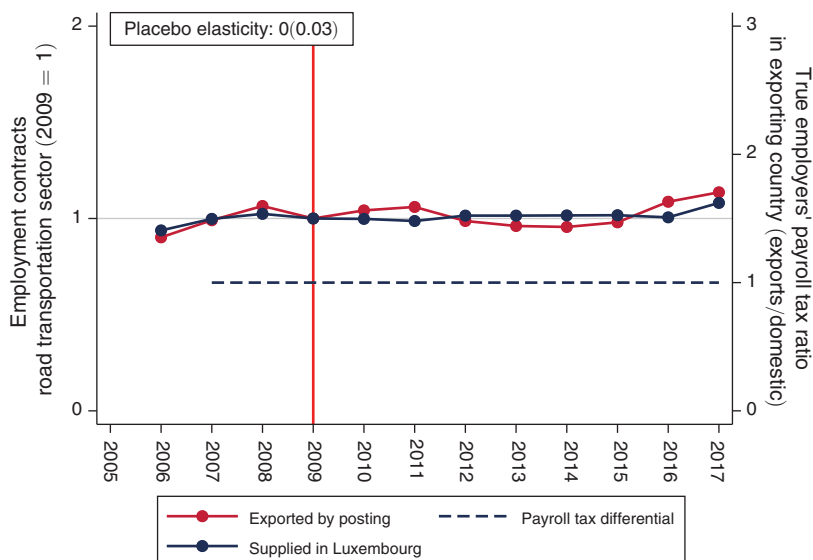


FIGURE 7. EFFECT OF IMPOSING DESTINATION-BASED TAXES ON LUXEMBOURGIAN EXPORTERS

Notes: This figure shows the effects of an EU regulation implemented in May 2010 that imposed destination-based payroll taxation on temporary employment agencies located in border regions posting workers in neighboring countries. The dashed line (and right y-axis) shows the evolution of the payroll tax rate ratio between exporters (treated) and firms that sell domestically (control). The top panel shows exports (treated, red series) versus domestic (control, blue series) contracts (measured in unit of workers) of temporary employment agencies in Luxembourg, before and after the reform (vertical red line). The bottom shows the same differential evolution for the road transportation sector, which was sheltered from the reform for 10 years due to a “transitory” sector-specific exemption. All series are normalized to one in the pre-reform year. The elasticity reported in the figure is computed with respect to the change in the statutory payroll tax rate. The implied elasticity with respect to the total labor cost is $-8.5(1.3)$ and is reported in panel B of Table 2.

Despite the absence of differential trends before the reform, the drop in posting exports in 2010 could be caused by a simultaneous shock affecting foreign but not domestic demand in the same year. This shock would only affect exports of posting services in the exposed sector, but not domestic supply of those services, and would violate the identifying assumption. Panel B of Figure 7 plots the yearly number of jobs in the road transportation sector in Luxembourg from 2006 to 2020 (normalized to one in 2009) posted abroad and supplied domestically. In this sector where posting is also prevalent, but that was not targeted by the EU regulation, postings did not decline after 2010 compared to domestic contracts. This rules out demand shocks abroad that would affect the foreign demand of services in the sheltered sector too. The implied placebo elasticity, reported in the graph, is zero and is statistically different from the elasticity estimated in the treated sector.⁴⁶ The absence of a response also confirms the lack of general equilibrium effects in nonexposed sectors: There was no reallocation of economic activity from the exposed to the sheltered sector following the reform.

Using the untreated sector allows me to directly relax the parallel trends assumption required for identification in panel A of Figure 7 by controlling for potential differences in pre-trends. Specifically, I can now estimate a gravity-based equation with the two sectors:

$$(10) \quad \log S_{ijts} = \alpha - \theta \log(c_{ijts}) + \gamma_{jt} + \gamma_{st} + u_{ijts},$$

where the labor cost c_{ijts} is instrumented with the three-way reform interaction $\mathbf{1}_{\{j=\text{abroad}\}} \times \mathbf{1}_{\{t \geq 2010\}} \times \mathbf{1}_{\{s=\text{temp. agency}\}}$. This specification compares services that are exported to those that are supplied domestically in the temporary employment sector versus the road transportation sector, before and after that payroll tax increase for exporters of temporary employment services only. The fixed effects now directly control for all time-varying shocks abroad (γ_{jt}) and sector-year shocks (γ_{st}) that could confound the estimates of θ . The elasticity with respect to the wage cost is large and statistically significant at the 1 percent level, with a coefficient of -5.35 (1.2) in the medium run.⁴⁷

Consistent with the evidence in Section IVA, Figure 7 shows that higher payroll taxes decreased firms' exports of labor-intensive services, with an elasticity above one. This demonstrates that differences in labor regulations shape trade flows between low and high-wage countries (e.g., as in the German reform) but also between otherwise similar high-wage countries (e.g., the Luxembourgian reform).⁴⁸

Interpretation.—A natural question is whether the responses in Figure 7 should be interpreted as relabeling responses, where previously posted workers become hired in the same position directly in the destination country. This would be conceptually

⁴⁶ I follow Kleven, Landais, and Saez (2013) and compute the placebo elasticity using the same tax differential as for eligible sectors in panel A of Figure 7.

⁴⁷ Another way to relax this parallel trends assumption is to investigate the effects of the reform on *imports* of posting services from Luxembourg in other countries. Supplemental Appendix Figure F.21 shows this alternative comparison in France, the main importer of Luxembourgian labor services before 2010.

⁴⁸ Figure 7 shows that payroll taxes shape the quantity of posting services traded across countries (the extensive margin). Supplemental Appendix G shows that payroll taxes also shape the length of those services (the intensive margin).

very similar to outsourcing, which is usually defined as employing the same worker for the same job at the same location, through temporary agencies instead of regular contracts (Goldschmidt and Schmieder 2017).⁴⁹

I investigate “relabeling” in Supplemental Appendix Figure F.22 using INSEE (2006–2017) data on French employment in temporary employment agencies and road transportation. If workers previously posted from Luxembourg are then hired in France and registered with the French social security system, they would appear in the French dataset after 2010. This is most likely to happen in French regions bordering Luxembourg, such as Moselle and Meurthe-et-Moselle, since most workers posted from Luxembourg to France reside in those *departements* and are posted there (Belkacem and Pigeron-Piroth 2016). I find no evidence that French employment increased in the exposed sector in French regions most exposed to the reform. The sharp, immediate drop in workers posted from Luxembourg in 2010 did not lead to the sudden hiring of those workers on the French side of the border. This is consistent with evidence from the gravity estimation (Supplemental Appendix Table D.7) and the German case study (Supplemental Appendix Figure I.29) and confirms limited substitution from posting to immigration in response to stricter labor regulations for posted workers.

Another way to test whether posting responses in Figure 7 only reflect relabeling is to study tax reforms in contexts where relabeling is less feasible. In Supplemental Appendix E, I study a large payroll tax cut in Belgium. The reform decreased posting flows to Belgium relative to a comparable destination country, with an elasticity of 3.8.⁵⁰ This confirms that trade in services is highly sensitive to labor taxes, including outside cross-border areas where posting contracts cannot be as easily used to relabel regular employment contracts.

V. Discussion

Across all empirical exercises, I find that trade in labor services is highly sensitive to labor costs. The estimates from the gravity model are larger than the steady-state correlations and are at the lower end of the reduced-form elasticities from the case studies. Several factors can explain these differences. First, the country-level measure of labor costs in the gravity estimation is not adjusted for sector-specific tax rates or minimum wages (as the one studied in the case studies), leading to attenuation bias. Second, the posting elasticity may vary across sectors and could be higher in activities such as temporary employment agencies. Third, it takes several years for posting flows to fully adjust to permanent changes in labor regulations. The case studies show a dynamic adjustment of posting flows to changes in labor regulations, with responses growing over time, while the year-to-year posting responses from the gravity estimation are smaller. These dynamics are similar to those documented by Boehm, Levchenko, and Pandalai-Nayar (2023) for trade in goods.

⁴⁹ If a German firm starts to hire workers through outsourcing, payroll taxes are still collected by Germany. With posting, this behavior implies that Germany will collect fewer social security contributions.

⁵⁰ I also show limited pass-through of the payroll tax reform on gross wages (Supplemental Appendix Figure E.15).

It is usually difficult to compare the magnitude of trade responses to (i) nonmonetary regulations and (ii) ad valorem tariffs, even though nontariff trade barriers represent more than 70 percent of all trade barriers globally (Kee, Nicita, and Olarreaga 2009; Niu et al. 2018). Furthermore, since services are not subject to tariffs, there is a lack of estimates for the trade elasticity in those sectors (Francois and Hoekman 2010). My setting is different because I can infer the euro-equivalent change in unit costs to produce services due to labor regulations. Because my gravity estimation is an empirical application of widely used trade models and relies on similar estimators and fixed effects as most other applications (Head and Mayer 2014), I can thus benchmark my estimates against those obtained in different contexts. I find that the trade elasticity for posting services is smaller than that for goods, which is usually around 5 (Head and Mayer 2014; Costinot, Donaldson, Komunjer 2012), though estimates relying on time series variation are smaller for goods too (Boehm, Levchenko, and Pandalai-Nayar 2023).⁵¹

Why would posting services be less responsive to changes in trade costs than trade in goods? One explanation is that it is harder for customers to substitute foreign services than foreign goods, possibly due to language barriers.⁵² Another possibility is that moving people across borders is more difficult than moving goods, making the posting elasticity closer to a migration elasticity. Monte, Redding, and Rossi-Hansberg (2018) estimate that the elasticity of commuting flows with respect to wages in the United States is 3. The elasticity of international migration flows of top earners in the European Union, which is more relevant to the context of posting, is around 1 (Kleven et al. 2020). Therefore, the posting elasticity is lower than the trade elasticity for goods but larger than the international migration elasticity, which is consistent with posting being temporary and intermediated by firms, making these flows more responsive to changes in migration costs. Finally, to interpret the coefficients in Table 1 as structural trade elasticities, the share of labor costs in unit costs must be one. If labor is not the only input, the elasticity for services needs to be appropriately scaled by the share of wages in the cost function, as in Eaton and Kortum (2002), and would be larger.⁵³

VI. Concluding Remarks: The “Polish Plumber” Parable

This paper has studied the effects of minimum wages and payroll taxes on international trade in labor-intensive services in Europe. The effects are identified using a number of reforms: minimum wage and payroll tax reforms within countries, the introduction of destination-based minimum wages for workers posted to Germany, and the introduction of destination-based payroll tax rates for workers posted from Luxembourg. The elasticity of posting flows to labor costs is above one and is even larger in some sectors. Hence, a country can successfully gain market shares in

⁵¹ Like Boehm, Levchenko, and Pandalai-Nayar (2023) for goods, I estimate a larger trade elasticity for services when only including multilateral resistance fixed effects but omitting bilateral fixed effects: The elasticity is -3.45 against the baseline of -1.2 .

⁵² This is a demand-side driven explanation, although the model that microfound the gravity equation naturally leads to a supply-driven interpretation of θ . An alternative model could be used to microfound a similar estimating equation with a different interpretation of the elasticity (Head and Mayer 2014).

⁵³ This could also rationalize the difference in estimates in Figure 4 and Figure 7 if the labor share differs in temporary employment agency sector and meat processing.

labor-intensive activities by reducing labor costs through payroll tax cuts or weaker labor standards. This implies that trade integration can limit the ability for countries to maintain independent labor market rules.

In his book, *Has Globalization Gone Too Far?*, Dani Rodrik (1997) illustrated the importance of considering social standards in the process of trade liberalization with a simple “parable.” In the first scenario, Honduran workers produce clothing in Honduras, following the labor laws of their home country. In the second scenario, Honduran workers temporarily travel to the United States to work in the clothing industry but are still governed by Honduras’ labor laws. From an economic standpoint, these two configurations are equivalent: The well-being of displaced US workers, Honduran workers, and firms’ shareholders are exactly the same. Hence, anyone basing their view on pure economic outcomes should support both policies. But, as Rodrik (1997, p. 34) puts it, “the vast majority of economists who have no difficulty with the outsourcing example would also accept that it is not good public policy to relax labor standards for migrant workers.”

While Rodrik (1997)’s parable was initially conceived as an extreme thought experiment to highlight the complex political considerations in the trade liberalization process, the posting policy has effectively brought this example to life in the real world. But many economic stakeholders shared Rodrik’s view that a commitment to free trade within the EU contradicted the enforcement of domestic minimum wages on foreign firms in the posting policy, and proposed removing this rule in 2005, through the so-called Bolkestein directive. The proposal became heavily unpopular in western Europe, as trade unions and left-wing political parties argued that it would lead to a “social dumping,” as service providers (such as Polish plumbers) from countries with lower standards would be able to offer their services in other EU countries without having to comply with the host country’s regulations. This “Polish plumber” polemic led to large protests in the European Union and is still viewed as the pivotal moment that marked the rejection of a globalization process based on competition in labor standards in Europe (Cagé and Piketty 2023). Like Rodrik’s allegory of Honduran migrants, the posting policy case study illustrates why the question of trade liberalization is hard to address separately from that of social convergence.

REFERENCES

- Agell, Jonas, and Per Lundborg. 1995. “Fair Wages in the Open Economy.” *Economica* 62 (247): 335–51.
- Albert, Christoph, and Joan Monras. 2022. “Immigration and Spatial Equilibrium: The Role of Expenditures in the Country of Origin.” *American Economic Review* 112 (11): 3763–802.
- Alesina, Alberto, and Roberto Perotti. 1994. “The Welfare State and Competitiveness.” NBER Working Paper 4810.
- Alfaro-Ureña, Alonso, Benjamin Faber, Cecile Gaubert, Isabela Manelici, and Jose P. Vasquez. 2021. “Responsible Sourcing? Theory and Evidence from Costa Rica.” Unpublished.
- Allen, Treb, and Costas Arkolakis. 2016. “Elements of Advanced International Trade.” Unpublished.
- Arkolakis, Costas, Arnaud Costinot, and Andrés Rodríguez-Clare. 2012. “New Trade Models, Same Old Gains?” *American Economic Review* 102 (1): 94–130.
- Auerbach, Alan J. 2010. *A Modern Corporate Tax*. Center for American Progress.
- Avi-Yonah, Reuven Shlomo, and Kimberly A. Clausing. 2007. “Reforming Corporate Taxation in a Global Economy: A Proposal to Adopt Formulary Apportionment.” Hamilton Project Discussion Paper 2007-08.

- Balassa, Bela.** 1964. "The Purchasing-Power Parity Doctrine: A Reappraisal." *Journal of Political Economy* 72 (6): 584–96.
- Barbiero, Omar, Emmanuel Farhi, Gita Gopinath, and Oleg Itskhoki.** 2019. "The Macroeconomics of Border Taxes." *NBER Macroeconomics Annual* 33: 395–457.
- Belkacem, Rachid, and Isabelle Pigeron-Piroth.** 2016. "L'interim Transfrontalier: Les Marges de L'emploi aux Marges des Territoires." *Revue Française de Socio-Économie* 17: 43–63.
- Benzarti, Youssef, and Alisa Tazhitdinova.** 2021. "Do Value-Added Taxes Affect International Trade Flows? Evidence from 30 Years of Tax Reforms." *American Economic Journal: Economic Policy* 13 (4): 469–89.
- Boehm, Christoph E., Andrei A. Levchenko, and Nitya Pandalai-Nayar.** 2023. "The Long and Short (Run) of Trade Elasticities." *American Economic Review* 113 (4): 861–905.
- Boudreau, Laura.** 2021. "Multinational Enforcement of Labor Law: Experimental Evidence from Bangladesh's Apparel Sector." Unpublished.
- Brecher, Richard A.** 1974. "Minimum Wage Rates and the Pure Theory of International Trade." *Quarterly Journal of Economics* 88 (1): 98–116.
- Bryan, Gharad, and Melanie Morten.** 2019. "The Aggregate Productivity Effects of Internal Migration: Evidence from Indonesia." *Journal of Political Economy* 127 (5): 2229–68.
- Cagé, Julia, and Thomas Piketty.** 2023. *Une Histoire du Conflit Politique: Elections et Inégalités Sociales en France, 1789–2022*. Seuil.
- Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer.** 2019. "The Effect of Minimum Wages on Low-Wage Jobs." *Quarterly Journal of Economics* 134 (3): 1405–54.
- CEPII.** 2007–2023. BACI: *International trade database at the product-level*. https://www.cepii.fr/CEPII/en/bdd_modele/bdd_modele_item.asp?id=37 (accessed January 2024).
- Chevreaux, Marine, and Remy Mathieu.** 2016. "Concurrence Sociale des Travailleurs Détachés en France: Fausses Évidences et Réalités." *Trésor-Éco Lettre* 171.
- Chor, Davin.** 2010. "Unpacking Sources of Comparative Advantage: A Quantitative Approach." *Journal of International Economics* 82 (2): 152–67.
- Costinot, Arnaud.** 2009. "On the Origins of Comparative Advantage." *Journal of International Economics* 77 (2): 255–64.
- Costinot, Arnaud, and Iván Werning.** 2019. "Lerner Symmetry: A Modern Treatment." *American Economic Review: Insights* 1 (1): 13–26.
- Costinot, Arnaud, Dave Donaldson, and Ivana Komunjer.** 2012. "What Goods Do Countries Trade? A Quantitative Exploration of Ricardo's Ideas." *Review of Economic Studies* 79 (2): 581–608.
- Cuñat, Alejandro, and Marc J. Melitz.** 2012. "Volatility, Labor Market Flexibility, and the Pattern of Comparative Advantage." *Journal of the European Economic Association* 10 (2): 225–54.
- De Wispelaere, Frederic, and Jozef Pacolet.** 2014. *Posting of Workers: Report on A1 Portable Documents Issued in 2012 and 2013*. European Commission; Directorate-General for Employment, Social Affairs and Inclusion. <https://hiva.kuleuven.be/nl/nieuws/docs/posting-of-workers-2012-and-2013.pdf>.
- De Wispelaere, Frederic, and Jozef Pacolet.** 2015. *Posting of Workers: Report on A1 Portable Documents Issued in 2014*. European Commission; Directorate-General for Employment, Social Affairs and Inclusion. <https://hiva.kuleuven.be/nl/nieuws/docs/posting-of-workers-2014.pdf>.
- De Wispelaere, Frederic, and Jozef Pacolet.** 2016. *Posting of Workers: Report on A1 Portable Documents Issued in 2015*. European Commission; Directorate-General for Employment, Social Affairs and Inclusion. https://books.google.com/books/about/Posting_of_Workers.html?id=ub9KzQEACAAJ.
- De Wispelaere, Frederic, and Jozef Pacolet.** 2017. *Posting of Workers: Report on A1 Portable Documents Issued in 2016*. European Commission; Directorate-General for Employment, Social Affairs and Inclusion. <https://op.europa.eu/en/publication-detail/-/publication/0357b7d3-49c7-11e8-be1d-01aa75ed71a1/language-en>.
- De Wispelaere, Frederic, and Jozef Pacolet.** 2018. *Posting of Workers: Report on A1 Portable Documents Issued in 2017*. European Commission; Directorate-General for Employment, Social Affairs and Inclusion. <https://op.europa.eu/en/publication-detail/-/publication/ba0fe609-e359-11ea-ad25-01aa75ed71a1/language-en>.
- De Wispelaere, Frederic, Lynn De Smedt, and Jozef Pacolet.** 2019. *Posting of Workers: Report on A1 Portable Documents Issued in 2019*. European Commission; Directorate-General for Employment, Social Affairs and Inclusion. <https://labourinstitute.eu/wp-content/uploads/2024/01/PD-A1-report-Reference-year-2019-1.pdf>.
- DGT-DARES.** 2016–2020. *Fichier Statistique*. SI-PSI DGT-DARES.
- Di Tella, Rafael, and Dani Rodrik.** 2020. "Labour Market Shocks and the Demand for Trade Protection: Evidence from Online Surveys." *Economic Journal* 130 (628): 1008–30.
- Direction Générale du Travail.** 2005–2016. *Déclarations de détachement (hors transport) des entreprises étrangères prestataires de services en France de 2005 à 2016*. Ministère du travail.

- Direction Générale du Travail.** 2016. *Analyse des Déclarations de Détachement des Entreprises Prestataires de Services en France en 2015*. Ministère du Travail, de l'Emploi et de la Santé. https://herveguichaoua.fr/IMG/pdf/2015_bilan_d_tachements_2015-2.pdf.
- Direction Générale du Travail.** 2019. *Analyse des Déclarations de Détachement des Entreprises Prestataires de Services en France en 2017*. Ministère du Travail, de l'Emploi et de la Santé. https://travail-emploi.gouv.fr/sites/travail-emploi/files/files-spip/pdf/rapport_2017_declarations_de_detachement.pdf.
- Direction Générale du Travail.** 2019. Plan national de lutte contre le travail illégal, Bilan 2016–2018 et priorités 2019–2021. Technical Report, Direction Générale du Travail. Ministère du Travail, de l'Emploi et de la Santé.
- Doelfs, Guntram.** 2012. “Werkverträge. 1, 02 Euro pro Schwein.” *Magazin Mitbestimmung* 12. <https://www.boeckler.de/de/magazin-mitbestimmung-2744-102-euro-pro-schwein-10682.htm>.
- Dragusanu, Raluca, Eduardo Montero, and Nathan Nunn.** 2022. “The Effects of Fair Trade Certification: Evidence from Coffee Producers in Costa Rica.” *Journal of the European Economic Association* 20 (4): 1743–90.
- Dustmann, Christian, Attila Lindner, Uta Schönberg, Matthias Umkehrer, and Philipp vom Berge.** 2022. “Reallocation Effects of the Minimum Wage.” *Quarterly Journal of Economics* 137 (1): 267–328.
- Eaton, Jonathan, and Samuel Kortum.** 2002. “Technology, Geography, and Trade.” *Econometrica* 70 (5): 1741–79.
- Eaton, Jonathan, and Samuel Kortum.** 2018. “Trade in Goods and Trade in Services.” In *World Trade Evolution*, edited by Lili Yan Ing and Miaojie Yu, 82–125. Routledge.
- Eaton, Jonathan, Samuel S. Kortum, and Sebastian Sotelo.** 2012. “International Trade: Linking Micro and Macro.” NBER Working Paper 17864.
- Erol, Serife, and Thorsten Schulten.** 2021. “Renewing Labour Relations in the German Meat Industry.” Institute of Economic and Social Research Report 61e.
- European Commission.** 2011. *Administrative Data Collection on E 101 Certificates Issued in 2008 and 2009*. Directorate-General for Employment, Social Affairs and Inclusion.
- European Commission.** 2012. *Posting of Workers in the European Union and EFTA Countries: Report on A1 Portable Documents Issued in 2010 and 2011*. Directorate-General for Employment, Social Affairs and Inclusion. https://www.google.com/url?sa=i&source=web&rct=j&url=https://ec.europa.eu/social/BlobServlet?docId%3D9675%26langId%3Den&ved=2ahUKEwiSxa-skMpAxXXIYkEHTL_Fl4QqYcPegQIABA7&opi=89978449&cd&psig=AOvVaw3PMxkh4oGXGasmrKHyrKkKu&ust=1756320095224000.
- Eurostat.** 2009–2018. *Employment by sex, age and economic activity*. DOI:10.2908/lfsq_egan2.
- Eurostat.** 2009–2018. *Labor costs annual data. 2009–2018*. doi:10.2908/LC_LCI_LEV.
- Eurostat.** 2009–2018. *Minimum wages*. DOI:10.2908/earn_mw_cur.
- Eurostat.** 2009–2018. *Population on 1 January by age group, sex and citizenship*. DOI:10.2908/migr_pop1ctz.
- Eurostat.** 2009–2018. *Purchasing power parities (PPPs), price level indices and real expenditures for ESA 2010 aggregates*. DOI:10.2908/PRC_PPP_IND.
- Fajgelbaum, Pablo D., Pinelopi K. Goldberg, Patrick J. Kennedy, and Amit K. Khandelwal.** 2020. “The Return to Protectionism.” *Quarterly Journal of Economics* 135 (1): 1–55.
- Farhi, Emmanuel, Gita Gopinath, and Oleg Itskhoki.** 2014. “Fiscal Devaluations.” *Review of Economic Studies* 81 (2): 725–60.
- Feldstein, Martin S., and Paul R. Krugman.** 1990. “International Trade Effects of Value-Added Taxation.” In *Taxation in the Global Economy*, edited by Assaf Razin and Joel Slemrod, 263–82. University of Chicago Press.
- Francois, Joseph, and Bernard Hoekman.** 2010. “Services Trade and Policy.” *Journal of Economic Literature* 48 (3): 642–92.
- Gan, Li, Manuel A. Hernandez, and Shuang Ma.** 2016. “The Higher Costs of Doing Business in China: Minimum Wages and Firms’ Export Behavior.” *Journal of International Economics* 100: 81–94.
- Gaulier, Guillaume, and Soledad Zignago.** 2010. “Baci: International Trade Database at the Product-Level (The 1994–2007 Version).” CEPII Working Paper 2010-23.
- Goldschmidt, Deborah, and Johannes F. Schmieder.** 2017. “The Rise of Domestic Outsourcing and the Evolution of the German Wage Structure.” *Quarterly Journal of Economics* 132 (3): 1165–217.
- Grienberger, Katharina, Markus Janser, and Florian Lehmer.** 2023. “The Occupational Panel for Germany.” *Jahrbücher für Nationalökonomie und Statistik* 243 (6): 711–24.
- Griffith, Rachel, James Hines, and Peter Birch Sørensen.** 2010. “International Capital Taxation.” In *Dimensions of Tax Design: The Mirrlees Review*, edited by Stuart Adam, Timothy Besley, Richard Blundell, Stephen Bond, Robert Chote, Malcolm Gammie, Paul Johnson, Gareth Myles, and James Poterba. Oxford University Press.

- Grossman, Gene M., Phillip McCalman, and Robert W. Staiger.** 2021. "The 'New' Economics of Trade Agreements: From Trade Liberalization to Regulatory Convergence?" *Econometrica* 89 (1): 215–49.
- Grossman, Gene M., and Esteban Rossi-Hansberg.** 2008. "Trading Tasks: A Simple Theory of Offshoring." *American Economic Review* 98 (5): 1978–97.
- Hanna, Rema.** 2010. "US Environmental Regulation and FDI: Evidence from a Panel of US-Based Multinational Firms." *American Economic Journal: Applied Economics* 2 (3): 158–89.
- Harrison, Ann, and Jason Scorse.** 2010. "Multinationals and Anti-sweatshop Activism." *American Economic Review* 100 (1): 247–73.
- Hassel, Anke, Jette Steen Knudsen, and Bettina Wagner.** 2016. "Winning the Battle or Losing the War: The Impact of European Integration on Labour Market Institutions in Germany and Denmark." *Journal of European Public Policy* 23 (8): 1218–39.
- Head, Keith, and Thierry Mayer.** 1948–2018. *Gravity Dataset: Goods*. <https://sites.google.com/site/hiegravity/home?authuser=0> (accessed October 2022).
- Head, Keith, and Thierry Mayer.** 2014. "Gravity Equations: Workhorse, Toolkit, and Cookbook." In *Handbook of International Economics*, Vol. 4, edited by Gita Gopinath, Elhanan Helpman, and Kenneth Rogoff, 131–95. Elsevier.
- Head, Keith, and Thierry Mayer.** 2021. "The United States of Europe: A Gravity Model Evaluation of the Four Freedoms." *Journal of Economic Perspectives* 35 (2): 23–48.
- Head, Keith, Thierry Mayer, and John Ries.** 2009. "How Remote Is the Offshoring Threat?" *European Economic Review* 53 (4): 429–44.
- Helpman, Elhanan, and Oleg Itskhoki.** 2010. "Labour Market Rigidities, Trade and Unemployment." *Review of Economic Studies* 77 (3): 1100–1137.
- IAB.** 2012–2018. *The IAB Occupational Panel*. <https://iab.de/en/daten/iab-occupational-panel/> (accessed March 2024).
- IGSS.** 2002–2020. *Luxembourg MicroData on Labour and Social Protection*.
- INSEE.** 2006–2017. *Base sur l'emploi localise: Emploi salarie en fin d'annee par departement et region de France*. <https://www.insee.fr/fr/statistiques/7727476?sommaire=4981513>.
- Kanbur, Ravi, and Michael Keen.** 1993. "Jeux sans Frontieres: Tax Competition and Tax Coordination When Countries Differ in Size." *American Economic Review* 83 (4): 877–92.
- Katz, Lawrence F., and Alan B. Krueger.** 2019. "The Rise and Nature of Alternative Work Arrangements in the United States, 1995–2015." *ILR Review* 72 (2): 382–416.
- Kee, Hiau Looi, Alessandro Nicita, and Marcelo Olarreaga.** 2009. "Estimating Trade Restrictiveness Indices." *Economic Journal* 119 (534): 172–99.
- Keen, Michael, and Kai Konrad.** 2013. "The Theory of International Tax Competition and Coordination." In *Handbook of Public Economics*, Vol. 5, edited by Alan J. Auerbach, Raj Chetty, Martin Feldstein, and Emmanuel Saez, 257–328. Elsevier.
- Kleven, Henrik, Camille Landais, Mathilde Muñoz, and Stefanie Stantcheva.** 2020. "Taxation and Migration: Evidence and Policy Implications." *Journal of Economic Perspectives* 34 (2): 119–42.
- Kleven, Henrik J., Camille Landais, and Emmanuel Saez.** 2013. "Taxation and International Migration of Superstars: Evidence from the European Football Market." *American Economic Review* 103 (5): 1892–924.
- Kleven, Henrik J., Camille Landais, Emmanuel Saez, and Esben A. Schultz.** 2014. "Migration and Wage Effects of Taxing Top Earners: Evidence from the Foreigners' Tax Scheme in Denmark." *Quarterly Journal of Economics* 129 (1): 333–78.
- Krueger, Alan B.** 1997. "International Labor Standards and Trade." In *Annual World Bank Conference on Development Economics 1996*, edited by Michael Bruno and Boris Pleskovic, 281–302. World Bank.
- Ku, Hyejin, Uta Schönberg, and Ragnhild C. Schreiner.** 2020. "Do Place-Based Tax Incentives Create Jobs?" *Journal of Public Economics* 191: 104105.
- Lerner, A. P.** 1936. "The Symmetry between Import and Export Taxes." *Economica* 3 (11): 306–13.
- Levchenko, Andrei A.** 2007. "Institutional Quality and International Trade." *Review of Economic Studies* 74 (3): 791–819.
- Linder, Staffan Burenstam.** 1961. *An Essay on Trade and Transformation*. Almqvist and Wiksell.
- Mah, Jai S.** 1997. "Core Labour Standards and Export Performance in Developing Countries." *World Economy* 20 (6): 773–85.
- Malgouyres, Clément, and Thierry Mayer.** 2018. "Exports and Labor Costs: Evidence from a French Policy." *Review of World Economics* 154 (3): 429–54.
- Manova, Kalina.** 2008. "Credit Constraints, Equity Market Liberalizations and International Trade." *Journal of International Economics* 76 (1): 33–47.

- Mense-Petermann, Ursula, Thomas Welskopp, and Anna Zaharieva, eds. 2022. *In Search of the Global Labor Market*. Brill.
- Monte, Ferdinando, Stephen J. Redding, and Esteban Rossi-Hansberg. 2018. "Commuting, Migration, and Local Employment Elasticities." *American Economic Review* 108 (12): 3855–90.
- Muñoz, Mathilde. 2024. "Trading Nontradables: The Implications of Europe's Job Posting Policy." *Quarterly Journal of Economics* 139 (1): 235–304.
- Muñoz, Mathilde. 2025. *Data and Code for: "International Trade Responses to Labor Market Regulations"*. Nashville, TN: American Economic Association; distributed by Inter-university Consortium for Political and Social Research, Ann Arbor, MI. <https://doi.org/10.3886/E229401V1>.
- Naidu, Suresh, Yaw Nyarko, and Shing-Yi Wang. 2016. "Monopsony Power in Migrant Labor Markets: Evidence from the United Arab Emirates." *Journal of Political Economy* 124 (6): 1735–92.
- Niu, Zhaohui, Chang Liu, Saileshsingh Gunessee, and Chris Milner. 2018. "Non-tariff and Overall Protection: Evidence across Countries and over Time." *Review of World Economics* 154 (4): 675–703.
- Nunn, Nathan. 2007. "Relationship-Specificity, Incomplete Contracts, and the Pattern of Trade." *Quarterly Journal of Economics* 122 (2): 569–600.
- OECD. 2005–2023. *The OECD-WTO Balanced Trade in Services Database*. OECD Publishing, Paris (accessed February 2024).
- OECD. 2009–2018. *Taxing Wages: Social security contributions (SSC)—employer rates*. OECD Publishing, Paris (accessed April 2022).
- ONSS. 2008–2019. *Office National de Securite Sociale: Fichier Statistique LIMOSA*.
- Ottaviano, Gianmarco I. P., Giovanni Peri, and Greg C. Wright. 2013. "Immigration, Offshoring, and American Jobs." *American Economic Review* 103 (5): 1925–59.
- Rodríguez-Clare, Andrés. 2010. "Offshoring in a Ricardian World." *American Economic Journal: Macroeconomics* 2 (2): 227–58.
- Rodrik, Dani. 1996. *Labor Standards in International Trade: Do They Matter and What Do We Do about Them?* Overseas Development Council.
- Rodrik, Dani. 1997. *Has Globalization Gone Too Far?* Institute for International Economics.
- Romalis, John. 2004. "Factor Proportions and the Structure of Commodity Trade." *American Economic Review* 94 (1): 67–97.
- Saez, Emmanuel, Benjamin Schoefer, and David Seim. 2019. "Payroll Taxes, Firm Behavior, and Rent Sharing: Evidence from a Young Workers' Tax Cut in Sweden." *American Economic Review* 109 (5): 1717–63.
- Samuelson, Paul A. 1964. "Theoretical Notes on Trade Problems." *Review of Economics and Statistics* 46 (2): 145–54.
- Shapiro, Joseph S., and Reed Walker. 2018. "Why Is Pollution from US Manufacturing Declining? The Roles of Environmental Regulation, Productivity, and Trade." *American Economic Review* 108 (12): 3814–54.
- Silva, J. M. C. Santos, and Silvana Tenreyro. 2006. "The Log of Gravity." *Review of Economics and Statistics* 88 (4): 641–58.
- Staiger, Robert W., and Alan O. Sykes. 2009. "International Trade and Domestic Regulation." NBER Working Paper 15541.
- Thum-Thysen, Anna, and Erik Canton. 2015. "Estimation of Service Sector Mark-Ups Determined by Structural Reform Indicators." *European Economy Economic Papers* 547.
- Tørsløv, Thomas R., Ludvig S. Wier, and Gabriel Zucman. 2018. "The Missing Profits of Nations." NBER Working Paper 24701.
- Vogel, Jonathan. 2007. "Institutions and Moral Hazard in Open Economies." *Journal of International Economics* 71 (2): 495–514.
- Wagner, Bettina, and Anke Hassel. 2016. "Posting, Subcontracting and Low-Wage Employment in the German Meat Industry." *Transfer: European Review of Labour and Research* 22 (2): 163–78.
- Wagner, Ines. 2018. *Workers without Borders: Posted Work and Precarity in the EU*. ILR Press.